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journal homepage: www.elsevier.com/locate/infoandorgScaling a tech-mediated community: Prefigurative enrolling atop evolving institutional logics[☆]Yunjung Pak^a, Timothy R. Hannigan^{b,*}, P. Devereaux Jennings^a, Michael Lounsbury^a^a Alberta School of Business, University of Alberta, Edmonton, Alberta T6G 2R6, Canada^b Telfer School of Management, University of Ottawa, Ottawa, Ontario K1N6B9, Canada

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ABSTRACT

How a community scales yet retains its fundamental character is a perennial puzzle - doubly so for a decentralized technology-mediated community. Based on our longitudinal case study of Ethereum, we argue that scaling can be done via a set of co-constitutive change processes called prefigurative enrolling. Our historical assessment of this case was assisted by structural topic modeling (STM) of blogs, crypto media, and news outlets. This revealed three phases in prefigurative enrolling. The first was around roadmaps for community imaginaries, shock events (a major hack), and reactive technology and governance adjustments. The second involved distributed experimentation, staking as anting-in, and anticipation of security and regulatory requirements in platform protocols. The third was the emotional reframing of field issues (such as sustainability), building a new tech-mediated community language and philosophy, and elaborating roles and interdependencies. These phases allowed for the democratic community logic to be gradually combined with a more market one in a blended configuration of responsible crypto finance. This paper contributes to work on technology community scaling, institutional co-constitution, and structural topic modeling.

1. Introduction

How a community scales, yet retains its fundamental character, is one of social science's perennial theoretical puzzles (Durkheim, 1893/2014; Simmel, 1898). A major concern of community members with scaling is that it will disrupt foundational values and practices (Rothschild-Whitt, 1979; Wellman, 1979), and potentially rupture the underlying community institutional logic (Lounsbury, 2007; Thornton, Ocasio, & Lounsbury, 2012). This is particularly the case when scaling leads to the community logic being counterposed with a market logic (Marquis & Lounsbury, 2007). Research has shown that a community's character may be preserved in scaling by leveraging its occupational members' moral imperative(s). Because these are linked to underlying logics, this can enable mobilizing for endogenous and deeper change (Anteby, 2010; Kim & Kim, 2022). In addition, to scale up effectively, cultural entrepreneurs may strategically decouple community beliefs from exchange practices in moral markets, like local foods (Hedberg & Lounsbury, 2020), then subsequently recouple community beliefs to practices. However, these entail uncertainty and hazards: both pathways face the risk of a diluted community mission, overly pluralistic value systems, and incompatible logic combinations

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* Corresponding author.

E-mail address: tim.hannigan@telfer.uottawa.ca (T.R. Hannigan).

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(Greenwood, Raynard, Kodeih, Micelotta, & Lounsbury, 2011; Kraatz & Block, 2017).

With the rise of digitization and technical occupations in institutional fields, this community scaling problem has become an even more profound challenge. When an organization scales, it can be difficult to sustain trust, norms, and behaviours that are foundational for collective capabilities and identity (Hargadon & Bechky, 2006). Technology has been shown to occasion decentralized organizing processes, in part by introducing uncertainty around scripts for interaction and disrupting traditional forms of authority (Barley, 1986). With online communities that are born digital, the radically decentralized nature of their organizational form (Massa, 2017) amplifies these issues. With minimal in-person interaction, members rely on digital traces of interaction behavior and authority is often flattened (O'Mahony & Ferraro, 2007). In order to scale effectively, technology occupations and firms – as communities – need new protocols, membership rules, roles, and scripts, but also must bridge to extant organizations and their groups (Howard-Grenville, Nelson, Earle, Haack, & Young, 2017; Kellogg, 2009). However, these new protocols, roles, and scripts are only as effective as the governance systems that legitimate and enforce them. For example, absent boundary enlargement and cultural balancing processes, they can lead to diluted missions, high pluralism and logic incompatibilities (Grodal & O'Mahony, 2017).

This points to the effortful governance of logics as a key concern in community scaling (Lounsbury & Glynn, 2019). There needs to be coherence in the community logic through scaling, or the community itself is at risk of fragmenting. For technology-mediated communities, there is a question of what role technology plays in this. Faik, Barrett, and Oborn (2020) have argued for a *co-constitutive process* approach where (i) logics focus attention in the perception of possible actions that a technology affords and (ii) through the use in practice, these technology affordances prompt actors to activate particular logics. This is a recursive and continuous cycle. When technology affordances scale via sensegiving, translation, and decoupling mechanisms (Faik et al., 2020), this can lead to social change through shifting the centrality and compatibility of logics. However, we still know little about how this co-constitutive process factors into the community scaling problem. Given that logics have a co-constitutive relationship with technology affordances, it becomes unclear how community governance endures. One potentially valuable approach is to examine a technology-mediated community longitudinally. Given the overarching power of logic configurations in stable fields (Zietsma, Groenewegen, Logue, & Hinings, 2017), it is important to consider technology-mediated communities embedded in nascent fields where logic configurations are still forming (Hannigan & Casasnovas, 2020). Here, skilled cultural entrepreneurs – “active agents capable of tapping cultural resources to construct firm identity and attain legitimacy” (Lounsbury & Glynn, 2019: 12) – are likely to matter in scaling processes. Therefore, we think it is important to examine the question of *how a technology-mediated community might scale using endogenous mechanisms grounded in field logics as a nascent field matures?*

To explore this question in depth, we rely upon a longitudinal case study of one well-known, technology-mediated community: Ethereum. This represents an alternative to the related technology-mediated community of Bitcoin (Brody & Couture, 2021; Dylan-Ennis, 2024; Hsieh & Vergne, 2023). Similar to preceding online communities (Faraj, Jarvenpaa, & Majchrzak, 2011), particularly those involved with open-source software production (O'Mahony & Ferraro, 2007; Von Krogh & Von Hippel, 2006), Ethereum is both a community of members and a technology mediating platform that supports digital applications (dApps) and a cryptocurrency (known as “Ether”). Like many decentralized blockchain communities, it has relied on its community of volunteers to create code; yet it has also depended on market forces to try to scale (DuPont, 2017; Massa, 2017; Seidel, 2018). Here, we examine how Ethereum, from its inception in 2016 through major technical changes in 2018 to 2022, navigated the technology-mediated community-scaling problem. It scaled its technology architecture and membership, while also managing to retain its community logic in the face of simultaneously embracing a market logic.

Our analysis of this longitudinal case is built on archival documents including news and blog sources, as well as online industry publications. Interviews, commentaries, and analyst assessments are all part of this rich corpus. The documents are assessed using historical analysis, combined with topic modeling (Villamor Martin, Kirsch, & Prieto-Nañez, 2023) – in particular, the technique of Structural Topic Modeling (STM) – in order to trace the framed field issues reflecting logics (Meyer & Höllerer, 2010) as well as responses by actors to those framed issues, both in discursive and practice terms (Croidieu & Kim, 2018). We show how the Ethereum community's technology architecture, formed at the outset based on an approach known in the blockchain field as “proof of work” (PoW). On the one hand, this reflected an organizing vision (Swanson & Ramiller, 1997) as a shared understanding among the community for how to make sense of and interpret the technology. On the other, the implementation of this vision in material terms meant that specific technology affordances that encoded roles and functions were embedded in this system (Leonardi, 2011). In symbolic terms, it was consonant with the democratic – almost anarchist – community logic that appealed to many of the initial participants of the blockchain field (Dylan-Ennis, Kavanagh, & Araujo, 2023; Seidel, 2018).

Due to a combination of internal strife and changes in the macro-institutional environment around sustainability, Ethereum engaged in a process of non-hierarchical, strategic change. In our findings, we call this process *prefigurative enrolling* and articulate how this was a means of overcoming inertia and the tech-community scaling problem. Here, the community adopted a new technology architecture based on an alternative approach known as “proof of stake” (PoS), thereby allowing the community to draw more on the surrounding market logic yet develop its own more responsible form of it. Our analysis shows how cultural entrepreneurship is both culturally embedded and materially grounded; logics are expressed in technology, and changing the configuration of logics happened alongside changing the technology architecture. By linking to the concept of prefiguration, we show how the early moments (Hannigan & Casasnovas, 2020) are ongoing experiments of possibility (Lounsbury & Glynn, 2019). To scale up possibilities, it is important to examine practices of framing and strategically manipulating logics (key concerns in cultural entrepreneurship), but also to actively enroll community members in a future by engaging with emotional process and outlining new roles.

2. Theory

2.1. Tech-mediated communities

The concept of community plays a central role in both organizational studies and sociology (Marquis, Lounsbury, & Greenwood, 2011). One definition sees communities as “collections of actors whose membership in the collective provides social and cultural resources that shape their action” (Almandoz, 2017: 191). Building such collectives is often a goal of technology-based groups precisely because of the individual and group dependence on technology as a mediator for key parts of their social exchange (Faik et al., 2020; Leonardi & Barley, 2010). Accordingly, online communities are often characterized by the affordances of the technology (Leonardi & Vaast, 2017; Massa, 2017). However, there is a tension between how these affordances can be used to create community but also can lead to a lack of stickiness with membership. In online communities, membership can be fluid, just like the movement of resources and ideas (Faraj et al., 2011). Open-source software communities demonstrate this as they often need to develop more social organization and mechanisms of governance than other forms of organization (Von Krogh & Von Hippel, 2006). Although many begin as loosely bounded founder-led communities, they endure as a form of peer production when they can retain commitments from volunteers who contribute to technical work (Feller, Finnegan, Fitzgerald, & Hayes, 2008) rather than disperse as cosmopolitans across multiple communities (Dahllander & Frederiksen, 2012). To coordinate the members without alienating them, these communities necessarily must address this tension and design governance systems (O'Mahony & Ferraro, 2007). Given the similar role that technology plays across these different types of groups and communities, as a short-hand, we nest them under the umbrella term “tech-mediated communities”.

Governance systems in tech-mediated communities evolve over time (O'Mahony & Ferraro, 2007). Once tech-mediated communities begin to build up their social relations and norms of interaction around their mediating technologies, they start to form a conception of authority (ibid), a collective identity, and culture (Bechky, 2003). These burgeoning communities work to preserve their identities and cultures through socialization and boundary maintenance mechanisms (Howard-Grenville et al., 2017). As part of those processes, they are likely to develop protocols as important forms of institutional “rules of the game” (Hsieh & Vergne, 2023) that detail techniques, coordinate workflows, and are attached to goals (O'Mahony & Bechky, 2008). These normative and regulative features collectively come to represent a community's governance system (O'Mahony & Ferraro, 2007; Scott, 2013), though there is debate as to which informal and formal elements are part of governance arrangements (Hinings, Logue, & Zietsma, 2017).

2.1.1. Tech-mediated community atop field logics

In tech-mediated communities, the identities, protocols, cultures, and boundary maintenance mechanisms are not created in a vacuum. Even in emerging fields, they draw from nearby field models and logics (Furnari, 2016; Hannigan & Casasnovas, 2020). Logics are “the socially constructed, historical pattern of material practices, assumptions, values, beliefs, and rules by which individuals produce and reproduce their material subsistence, organize time and space, and provide meaning to their social reality” (Thornton & Ocasio, 1999: 804). Being deeply and widely held, institutional logics help to constitute a field of activity and thought in which those actors engage and help to reproduce.

This embeddedness with logics in fields is pronounced with a new wave of tech-mediated communities using blockchain for their attachment to the ideal of decentralized organization (Seidel, 2018) as well as imaginaries that animate member involvement (Brody & Couture, 2021). In a tech-mediated community, this can be illustrated with the concept of sociotechnical imaginaries, as “collectively held, institutionally stabilized, and publicly performed visions of desirable futures” (Jasanoff & Kim, 2015: 153). Shared understandings of possibilities with social order occur with and through technologies. For example, decentralized autonomous organizations or “DAOs” (Zachariadis, Hileman, & Scott, 2019; Zavolokina, Bauer-Hänsel, Hacker, & Schwabe, 2024) that are enabled by Ethereum are dependent on experience with democratic ideals (Dylan-Ennis et al., 2023; Seidel, 2018). Equally, technology community market exchanges are influenced by the forms of exchange predominate in an era. This raises a question of how communities are embedded within configurations of logics. Today, finance capitalism and its notions of fungible and mobile capital controlled via financial networks of investors is predominant in many markets (Davis & Kim, 2015). It is not surprising, therefore, to see many fields impacted by market logics associated with capital accumulation and limited fiduciary responsibility (Meyer & Höllerer, 2010).

2.2. Tech-mediated community scaling and community logic preservation

Tech-mediated communities are typically formed with the explicit goal of *scaling up* the technology that they produce (Howard-Grenville et al., 2017; Massa, 2017; O'Mahony & Ferraro, 2007). To scale up also means to spread the network, organization, or new occupation(s) associated with the community (Powell, White, Koput, & Owen-Smith, 2005), as well as the underlying associated logic (Hedberg & Lounsbury, 2020). This scaling occurs in a wider field, where extant methods of organizing may (or may not) be amenable to such scaling.

One classic scaling method for tech-mediated communities is diffusion (replication) of the community form. We see this in instrument manufacturers (Audia, Freeman, & Reynolds, 2006), telephone systems (Barnett, 1990), and techniques like total quality management (Westphal, Gulati, & Shortell, 1997). However, replication of this form is always incomplete and imperfect. Software development rules and occupations end up with many exceptions and enclaves of practice. Another scaling up method identified in the literature has to do with mobilization that entails boundary spanning and membership enlargement. We see this with open-source software and science community enlargements (Grodal, 2018; Howard-Grenville et al., 2017; O'Mahony & Bechky, 2008). Nevertheless, cultural entrepreneurship during innovation and mobilization is likely to create rival applications and forms that split the field,

as studied in the cases of mutual funds (Lounsbury, 2007) and TV streaming (Gurses & Ozcan, 2015).

Taking the materiality of technology more seriously in the scaling of social change, Faik et al. (2020) argued for more attention to be paid to how a community's technology and its affordances interact with the beliefs and practices of a community's logic. While prevailing bundles of key beliefs and practices that comprise a logic are likely to imprint most organizations in a given institutional field (Greenwood, Suddaby, & Hinings, 2002; Scott, 2013), the technologies used by organizations and their products consumed simultaneously provide a *material* anchor for those logics, and shape how such logics are instantiated and evolve (Jones, Boxenbaum, & Anthony, 2013; Leonardi, 2011). The affordances – “the range of functions and constraints that an object provides for, and places upon, structurally situated subjects” (Davis & Chouinard, 2016: 241) – shape the meaning and uses of the technology. Affordances are perceived and understood by users in practice. Logics focus attention on particular affordances (Faik et al., 2020), but this is not deterministic. The technologies, in other words, are not infinitely malleable under logics and field practice.

We seek to advance our understanding of how a tech-mediated community scales by probing how skilled cultural entrepreneurs and community members manage the relationship of technology affordances and prevailing logics in order to scale without fully sacrificing the character and identity of their community (Lounsbury & Glynn, 2019). As the extant literature has suggested, for most fields, scaling entails the growing prominence of market logics often at the expense of the charisma of community logics that animated initial field formation and development (Wry, Lounsbury, & Glynn, 2011).

2.2.1. Prefigurative enrolling

To address these issues with nascent fields or ones undergoing a more radical regime shift, as a start, we think it is important to begin with a consideration of how a community enacts futures (Augustine, Soderstrom, Milner, & Weber, 2019). In sociotechnical imaginaries (Jasanoff & Kim, 2015), visions of desirable futures stabilize and become collectively held at macro-level. In a nascent field or radically changing community, some members may not want to wait for such consensus and engage in direct action to explore alternative futures. The concept of “prefigurative organizing” (Reinecke, 2018) captures how community members experiment and break down the temporal distinction between the means of the present and the desired ends, so that the future gets practiced in the present. In their study on Ethereum, Dylan-Ennis et al. (2023) combined prefiguration and imaginaries, developing the concept of “prefigurative imaginaries”. In their paper, they discussed three imaginaries: (i) *technical imaginary*: a decentralized world computer, (ii) *an economic imaginary*: a productive form of money, and (iii) *a public imaginary*: public goods. They argued that these three overlapping imaginaries were held together by prefiguration, where experimentation and multiplicity characterized the Ethereum community. These imaginaries were constantly being tested and remade in practice with technology affordances (i.e. modifying the protocol and using smart contracts).

However, in considering how a tech-mediated community scales, it is unclear from these concepts how certain imaginaries gain prominence and mobilize support. Prefigurative imaginaries shows how alternative futures get articulated as prototypes in the Ethereum protocol, but there is a gap in our understanding of how people in tech-mediated communities can mobilize support. To scale up experiments and transform the organization material nature of such a tech-mediated community, it is necessary to engage more directly with a community logic (Thornton et al., 2012). In keeping with a co-constitutive approach (e.g., Faik et al., 2020), we argue that cultural entrepreneurs in a tech-mediated community are likely to engage wider audiences and to get them to commit to participating in collective action to make imaginaries real. Theoretically, one would normally argue that framing would be sufficient to convince members to back various proposals (Benford & Snow, 2000; Litrico & David, 2017). However, in the case of protocols for governing technology affordances, framing needs to be *combined with* protocols and material configurations of the technology (Grodal, 2018; Gurses & Ozcan, 2015; Howard-Grenville et al., 2017). Therefore, we focus on the technical imaginary of Ethereum as a decentralized “neutral world computer” (Dylan-Ennis et al., 2023) and its materialization both in affordances as protocols and as a technological architecture. The science and technique of the protocols, the beliefs and training of the professionals or skilled occupation members, and the material processes of the organizations where they are created need to be engaged with along with the wider ideas and beliefs of the field (Jones et al., 2013).

To do this, we think that a form of mobilization of these local field members to engage with the technology occurs via the use of prefigurative imaginaries, and that in the early stages of a new field, or when it transforms itself, this can be a general process - what we call “*prefigurative enrolling*”. We define prefigurative enrolling as *a set of processes by which cultural entrepreneurs are able to encourage wider audiences to assist in the collective materialization of a prefigurative imaginary*. The remainder of this paper explores prefigurative enrolling as part of Ethereum's unexpected, yet successful shift away from its prior proof-of-work technology and protocols to one based on the proof-of-stake approach while retaining key elements of its founding community logic and governance principles.

3. Methods

3.1. Empirical context

Ethereum and the blockchain technology on which it is based have been employed in prior work to examine technology affordances and community governance, making it a representative case for study (DuPont, 2017; Dylan-Ennis et al., 2023; Seidel & Greve, 2017). The Ethereum platform, which was created formally in 2015, was a follow-on effort from Bitcoin. In 2008, Bitcoin was created as both the first mainstream blockchain platform and cryptocurrency. On a technological and organizational basis, this was centered on mining (building) blocks of code to solve cryptographic mathematical problems and then capturing the value of those blocks through peer evaluations in the community (Hsieh & Vergne, 2023). This “block mining” innovation allowed for cryptocurrency generation to be decentralized, while relying on standardized underlying principles for coordination (Zachariadis et al., 2019). Unlike many

cryptocurrencies early on, Ethereum was also developed around the notion of smart contracting (Murray, Kuban, Josefy, & Anderson, 2021). Smart contracts are self-executing agreements based on contract terms written into code and stored on the blockchain. This extended the range of uses for blockchains to applications beyond finance.

Bitcoin was founded based on social imaginaries around alternative financial system (Swartz, 2018) that organized values, beliefs, and practices around a market logic (Thornton et al., 2012) (see Table 1, column 1). Ethereum was originally pitched as a decentralized “neutral, general-purpose world computer” (Buterin, 2014) that would run different applications (dApps) and in addition to a crypto currency (Dylan-Ennis et al., 2023). It also exhibited the values of open participation and community-decision making that correspond with a community logic (Lounsbury, 2007).

Ethereum's governance process normatively required that members of the community submit written Ethereum Improvement Proposals (EIPs) that detailed new functionality and behaviours for the protocols and technology. For example, the proposal EIP-721 (the “Non-Fungible Token Standard”, Ethereum Foundation, 2025) outlined and implemented the functionality of NFTs within smart contracts. This helped to establish a fixed digital asset that could be exchanged. After EIPs were pitched, the community subsequently discussed and made consensus on these proposals (Zachariadis et al., 2019). A successfully implemented EIP meant changing the Ethereum protocol and technology architecture. In order to implement this, the community needed to spawn a new base of source code – known as a “hard fork” (Hsieh & Vergne, 2023; Zachariadis et al., 2019). When the community migrated to using the new fork, this was known as a forking event on the platform. It represented a major event for the Ethereum community, and was often given a memorable label, such as a city like “The London Fork”. In this process, some EIPs were successfully implemented without issue, but the community often rejected them, even after some were seemingly approved after the formal EIP processes. This reflects the complex overlap between formal and informal elements (Hinings et al., 2017) in Ethereum's governance system.

For Ethereum, in 2015, it appeared the community had been stabilized, but in 2016 it was faced with a crisis in the form of a highly visible successful hack of the platform. This could have destroyed the initial community, but did not (Hannigan, Pak, & Jennings, 2022; Zachariadis et al., 2019). This time point then is the starting point of our in-depth longitudinal case analysis on scaling (see Fig. 1). Following it, there were more gradual, if still challenging, transition through a series of events from 2016 to 2022, which led to Proof of Work to Proof of Stake (see Supplementary Table 1 for a comparison of their differences). These events included more than one set of EIPs – an important “Beacon chain” test network (testnet) with the main network (mainnet) and only solidified in the fork entitled “The Merge” the transition of the Ethereum community to PoS (Ethereum Foundation, 2024a). The acceptance of POS is the end of our longitudinal case period.

3.1.1. From a democratic community logic to a blended responsible finance market logic

According to our preliminary historical analysis and others' research on the field (Dylan-Ennis et al., 2023; Jasanoff, 2015), the transformation of the architecture from PoW to PoS was accompanied by a shift in prominence of the logic governing the Ethereum community. Table 1 details key differences between these logics on which Ethereum relied. Each was a local instantiation of a broader field logic (Hedberg & Lounsbury, 2020; Lee & Lounsbury, 2015). At its inception, the community logic embraced direct democracy (Dylan-Ennis et al., 2023; Seidel, 2018). That logic relied on community members' creativity and contributions to source code as the basis of value. The work and value were directly transferable through the cryptocurrency Ether, as members were often incentivized by the protocol to engage in various tasks such as maintaining the public ledger. As a resource, Ether was also traded within the community and used in collective organizing efforts such as DAOs or initial coin offerings (ICOs) (Hsieh & Vergne, 2023; Zachariadis et al., 2019). A separate imaginary around Ether as “productive money” (as known as “crypto-fuel”) that corresponds with a market logic (Thornton et al., 2012) was present in the early moments (Hannigan & Casasnovas, 2020) of Ethereum (Buterin, 2014), but this also

Table 1
Ethereum's logics from the blockchain crypto field.

Logic Elements	Crypto Finance Market Logic	Community Democratic Community Logic	Responsible Finance Market Logic
Community Ideology	Unregulated Financial Market	Direct Democracy with Founder Guidance	Adjudicated Financial Market with Community Oversight
Source of Value	Profit-seeking through Individuals' Free Market Participation	Community Charter Commitments; Collective Participation in Building Shared Infrastructure	Profit-seeking Constrained by Community Obligations (and Sustainability Principles)
Source of Legitimacy	Market Efficiency and Censorship Resistance; Digital Gold Vision	Democratic Participation and Technical Innovation; World Computer Vision	Sustainable Growth and Responsible Innovation (and Regulatory Compliance)
Focus of Attention	Building an Unregulated Alternative to State-Centered Financial System	Building Decentralized Application Platform and Community	Building a Profitable Adjudicated System
Technological Protocol	Maintain Public Ledger Through Single Layer	Maintain Public Ledger Through Single Layer	Maintaining Legitimacy with Regulators
Technology Architecture	Proof of Work	Proof of Work	Maintain Public Ledger Through Two Layers
Coordination Principles	Market Competition among Atomized Miners	Community Consensus through Collective Deliberation	Coordinated Interdependence between Validators, Developers, and Foundation
Production System	Transactional Exchange	Relationally Bounded Transactional Exchange	Balanced Relational - Transactional Exchange
Supporting Organizational Form	Atomistic Independent Miners and Developers	Decentralized Autonomous Organizations (DAOs)	Moderately Decentralized Platform; Validator Networks

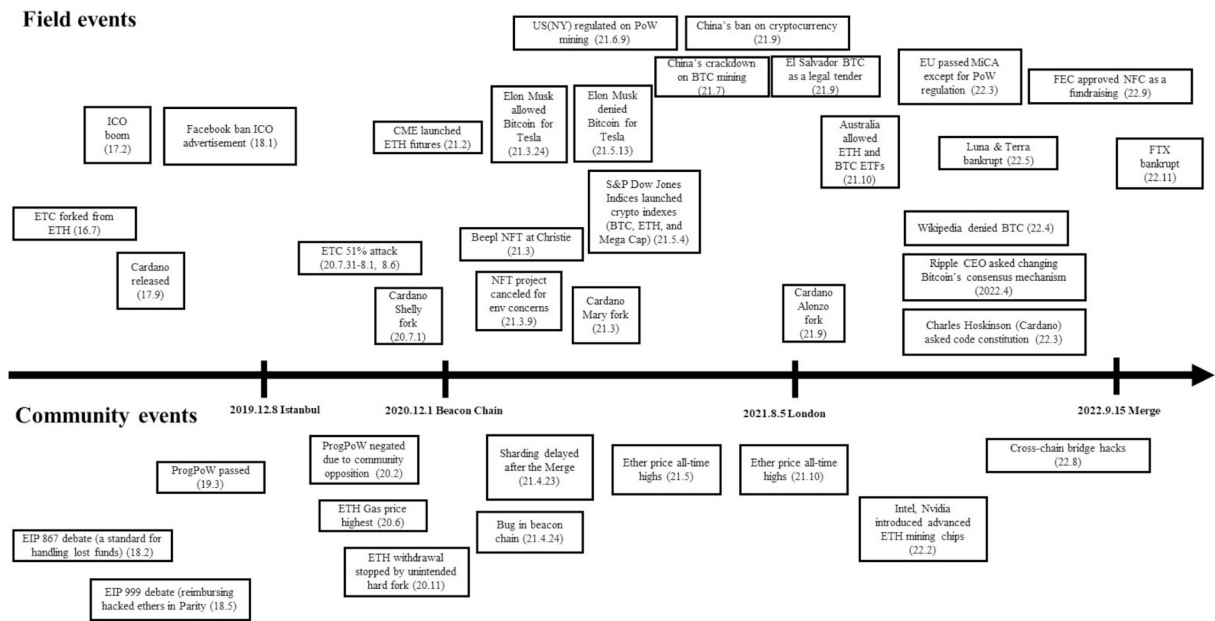


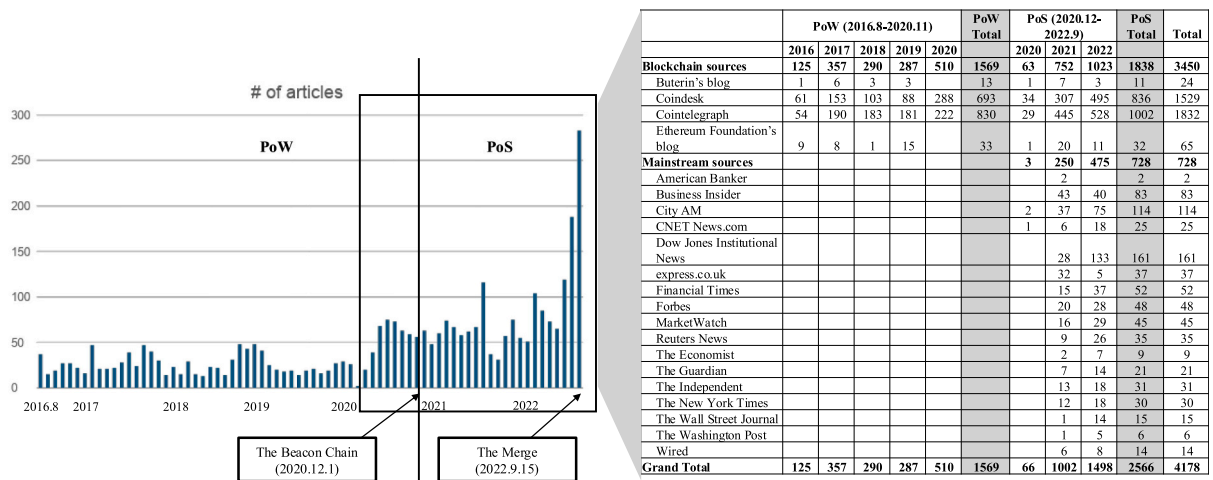
Fig. 1. A timeline of Ethereum events and issues, 2016–22.

appeared alongside other imaginaries such as those around the platform supporting public goods (Dylan-Ennis et al., 2023). However, for most of the early years of Ethereum, the community logic was foregrounded, and the market logic was backgrounded.

This democratic community logic of Ethereum was distinct from the logic followed by Bitcoin (Brody & Couture, 2021; Kinney, 2020). Studies on Bitcoin's culture point to its origins as a blend of the crypto-anarchism and cypherpunk movements (Dylan-Ennis et al., 2023; Hayes, 2019). Swartz (2018) examined how imaginaries within Bitcoin's culture even had internal contradictions. For example, the “digital metallism” and “infrastructure mutualism” that positioned Bitcoin as both a commodity (like digital gold) and as payment infrastructure (Swartz, 2018). Although both imaginaries pulled in differing directions of speculation and stability, they both corresponded with a market logic. This can be characterized in the first column of Table 1 as a form of crypto finance market logic. Nevertheless, that logic- being in the technology field - did influence Ethereum as the platform tried to scale (Yan, Almandoz, & Ferraro, 2021). Below, when focusing on the evolutions of Ethereum's technology community's governance features, we touch upon these parallel, evolving logics. We present evidence that by 2022 there was a form of blending of the two local logics underpinning the technology community (Table 1).

Table 2

Source of data and periodization of the transition.



3.1.2. Data collection to capture technology evolution and enrolling

To investigate the evolving governance mechanisms as the Ethereum community tried to scale, including some of the underpinning logics and core technology (see Fig. 1), we used a mix of archival historical data and media data on the firm and its local field (Croidieu & Kim, 2018; Grodal, 2018; Gurses & Ozcan, 2015; Litrico & David, 2017). Technology affordance scholars, in particular, have a tradition of gathering archival and discursive data around innovations garnered from participant observations, interviews, and transcripts in order to surface key points of attention, events, relational interpretations, and paths of development (Garud & Rappa, 1994; Kennedy, 2008; Leonardi, 2011; Orlikowski & Barley, 2001). The features in the discourse hook onto and reflect technology artifacts, networks, and platforms. In this way one can distinguish in the discourse between more social and more material features to which some data refer. Institutionally oriented researchers have also relied on these social versus material features referred to in discursive data, as we see in the historical analysis of ham operators of radio technology and its technology affordances versus behavioral correlates in the pre- and post-WWI era (Croidieu & Kim, 2018) and in the study of Pay TV's rise in the 1950s (Gurses & Ozcan, 2015). Our coding protocol, therefore, distinguishes between more political issues and social relations versus activities involving objectified artifacts, such as coding, platform structure and forking, that are classified as more material.

The discursive corpus that we created to trace Ethereum's scaling efforts had, at its core, the community's own internal discussions. Particularly Vitalik Buterin's (co-founder of Ethereum) and the Ethereum Foundation's blogs. However, given the width of observation window, we also relied on blockchain commentators and well-known media outlets. This included some of the regular intermediaries (technology field bloggers, particular outlets, writers) who were part of the interpretive and transitional processes (Zietsma et al., 2017). These were found on recognized blockchain outlets, such as *Coindesk* and *Cointelegraph*, and in mainstream publications such as

Table 3
Glossary of domain-specific terms.

Term	Definition
51% Attack	A security vulnerability where an actor controlling >50% of mining power or staked assets can manipulate transactions or execute double-spending (Ethereum Foundation, 2024b).
Beacon Chain	The Proof of Stake consensus chain launched in December 2020 that ran in parallel to Ethereum mainnet before being merged in September 2022 (Ethereum Foundation, 2024b).
Burning the Base Fee	A mechanism introduced in the London Fork (2021) where the base portion of transaction fees is permanently removed from circulation, reducing Ether supply and mitigating 51% attack incentives.
Consensus Layer	The Proof of Stake component of Ethereum (formerly Eth2/Beacon Chain) responsible for validator attestations, block proposals, and maintaining network consensus (Ethereum Foundation, 2022).
DAO	(Decentralized Autonomous Organization) Organizations governed by smart-contract rules with token-holder decisions rather than traditional management (DuPont, 2017).
DAO Hack	The 2016 security breach in which \$60 M in Ether was stolen from The DAO, Ethereum's first major smart contract experiment. This crisis led to a contentious hard fork splitting Ethereum into Ethereum (ETH) and Ethereum Classic (ETC) (DuPont, 2017).
Digital Metallism	A social imaginary viewing cryptocurrency as a digital analogue to precious metals as scarce, valuable, and independent of institutional backing (Swartz, 2018).
Ether	Ethereum's native cryptocurrency, used for transaction fees, gas, and staking.
Ethereum Virtual Machine (EVM)	Ethereum's computational engine that executes smart contracts and runs decentralized applications, enabling developers to code in multiple languages.
Execution Layer	The component of Ethereum (formerly Eth1/mainnet) responsible for executing smart contracts, processing transactions, and running the EVM (Ethereum Foundation, 2022).
FTX Crisis	Collapse of the FTX exchange due to fraud (November 2022), leading to market instability and heightened regulatory scrutiny across the cryptocurrency sector.
Gas Fees	Transaction fees paid to compensate validators/miners for computational resources required to process transactions and execute smart contracts.
Hard Fork	A permanent, non-backward-compatible modification to a blockchain's rules that diverges from existing blocks, often splitting the community into separate blockchain platforms (Liu et al., 2022).
Infrastructure Mutualism	A social imaginary highlighting collective ownership of shared digital infrastructure, portraying blockchain platforms as commons designed for mutual benefit (Dylan-Ennis et al., 2023).
Mainnet	The primary Ethereum network where real transactions take place and actual value is exchanged (the execution layer post-Merge) (Ethereum Foundation, 2022).
Mining	In Proof of Work systems, the use of computational power to validate transactions and add blocks, receiving rewards and fees.
Network State	A concept suggested by Balaji Srinivasan describing highly aligned online communities capable of collective action, crowdfunding territory, and seeking recognition, framing Ethereum as a sovereign digital community.
NFTs (Non-Fungible Tokens)	Unique digital assets stored on the blockchain that certify ownership of specific items (usually art or virtual real estate). Unlike cryptocurrencies, each NFT is distinct and non-interchangeable (Zachariadis et al., 2019).
Proof of Stake (PoS)	A consensus mechanism selecting validators based on their staked assets rather than computational power; substantially more energy-efficient (Buterin, 2014; Beck et al., 2018).
Proof of Work (PoW)	A consensus mechanism in which miners solve computational puzzles to validate blocks; highly energy-intensive (Beck et al., 2018).
Smart Contracts	Self-executing agreements with terms written into code and stored on the blockchain, automatically executing when conditions are met (Murray et al., 2021).
Staking (Staked Contracts)	Locking up cryptocurrency (Ether) as collateral to participate in blockchain validation activities and earn rewards in Proof of Stake systems (Beck et al., 2018).
The Merge	Ethereum's transition from Proof of Work to Proof of Stake in September 2022, unifying the mainnet execution layer with the Beacon Chain consensus layer.
Zombie Networks	Term used by Ethereum community members to describe rival blockchain platforms or forks (e.g., Ethereum Classic) perceived as outdated or lacking long-term viability.

The New York Times, *The Wall Street Journal*, *Forbes*, *Financial Times*, and *The Washington Post*. We gathered these articles using *Factiva*. Our search query was around “Ethereum” in conjunction with “fork,” “forking,” “the Merge,” “proof of work,” or “proof of stake.” Search queries from well-placed outlets are a standard method for creating textual corpora (Klein & Amis, 2021). Our media and historical corpus consisted of 4178 articles (see Table 2). That table is divided into blog and recognized media sources, and the attending figure shows the pre- and post-major data frequencies around the “merge” forking event. Together, these data artifacts, we believe, indicate a reasonable coverage of the evolving technology-focused discourse around Ethereum's scaling in its local field.

Given the incidences of domain specific language of this empirical setting, Table 3 provides additional definitions and explanations. Ethereum is a tech-mediated community that reflects on its ongoing activities and consciously encodes certain colloquial language into its cultural toolkit (Lounsbury & Glynn, 2019). This is not limited to labels about its technology, they also reflect on underlying philosophies, mechanisms for practices, and historical events.

3.2. Data analysis

To assess these data and refine our interpretations of their meanings and the processes of transition they captured, we took an abductive, computational approach (Hannigan et al., 2019; Timmermans & Tavory, 2012). This is a relatively new approach to coding discursive data, being dependent on content focused interpretation, including some axial coding, at key stages. It is fundamentally driven by quantitative assessment of large numbers of documents and words using partially automated (supervised) methods (Mohr et al., 2020). In this article, we use a version of topic modeling known as structural topic modeling or “STM” (Roberts, Stewart, & Tingley, 2015) in an abductive manner (Villamor Martin et al., 2023). This allows analysts to capture the change in identified first-order topics and their second-order themes systematically by some key covariates, such as time or type of actor.

We employed the evolving standard procedures for rendering our topic models (Grimmer, Roberts, & Stewart, 2022; Hannigan et al., 2019; Schmiedel, Müller, & Vom Brocke, 2019). In brief, in the first step the number of interpretable topics was based on

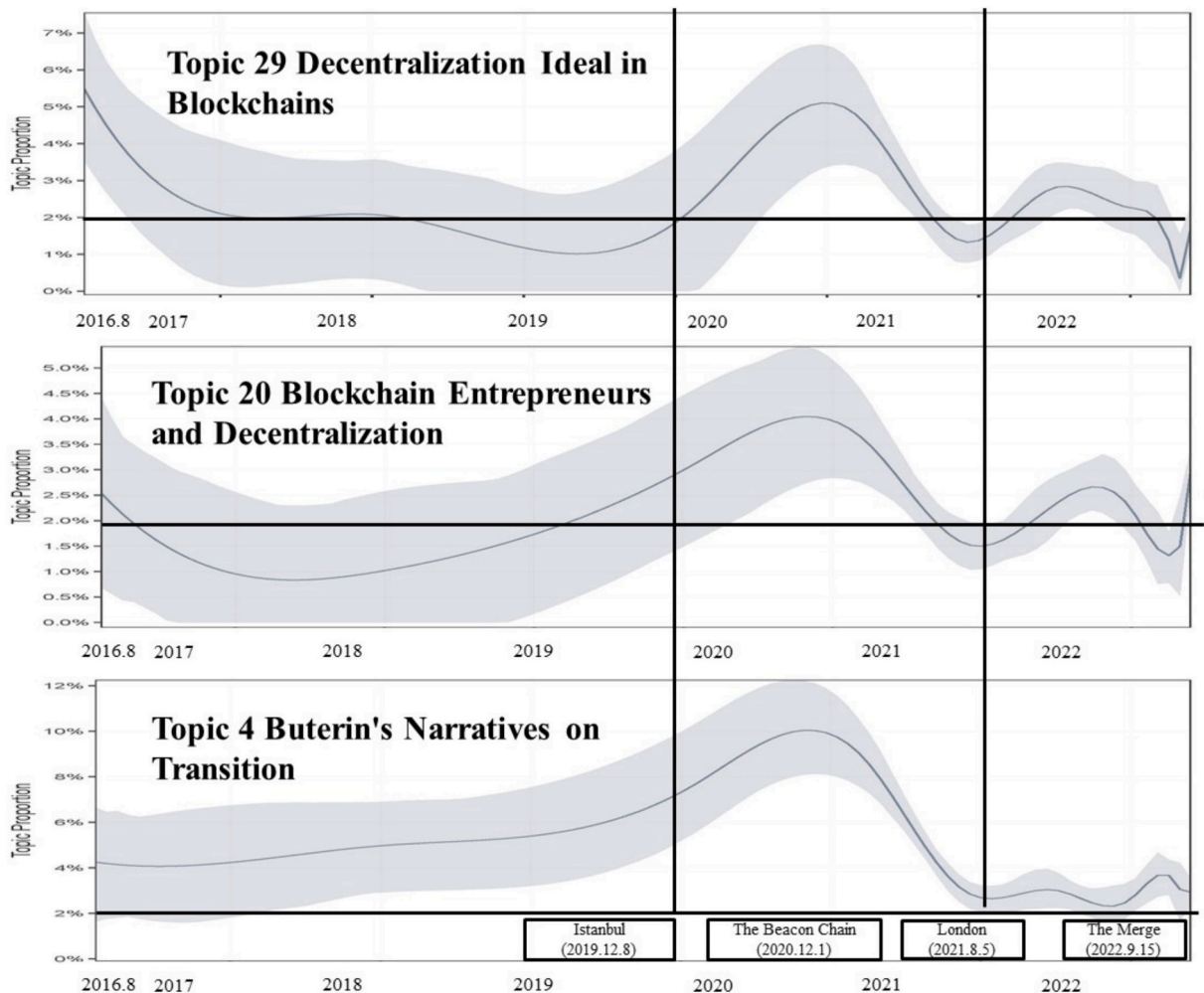


Fig. 2. The rise and fall of discourse around enrolling-related topics and forks.

assessing a short list of models that made up the “frontier” of jointly maximizing the diagnostic measures of semantic coherence and exclusivity (Roberts et al., 2015), along with interpretability (DiMaggio, Nag, & Blei, 2013). In the second step, the labeling of the 40 identified topics was conducted based on keywords and documents and preliminary focal topic candidates identified (see Supplementary Table A2); in the third, the 40 topics were collapsed into second-order groups, some more evident earlier versus later as the platform scaled (Supp. Table A3); in the fourth, key (focal) topics central to the community debates around the transition were theoretically identified, and changes in attention to them tracked and matched with material blockchain forks by Ethereum. In the fifth step, the top articles relating the focal topics and correlated one were classified and thoroughly read (see Supp. Table A4). We also generated a list of key actors based on these documents to aid our interpretation (Supp. Table A5). The more social and material discourse tied to these actors around evolving focal topics, like blockchain decentralization, allowed us to uncover key mechanisms associated with these (Bohn & Rogge, 2022). The sixth step was to build up a narrative around the evolving platform and lay the groundwork for final step of building a theoretical artifact from it (Hannigan et al., 2019), in this case a process model.

4. Findings

In our structural topic modeling analyses, the three focal topics on which we concentrated to which other topics were linked were the *decentralization ideal* (topic 29); *cultural entrepreneurship and technology change* (topic 20), and *the founder's narratives* (topic 4). Fig. 2 displays the rise and fall of these focal topics from 2016 to 2022. If one examines Fig. 1 of key events around Ethereum and its local field, it is evident that the rise and fall in Fig. 2 can be periodized based on the key platform forks: the two substantive forks (Beacon Chain fork and Merge fork) and two more incremental ones (Istanbul fork and London fork). These forks and the phases they create are depicted in Fig. 2 – and in subsequent figures and tables – as vertical lines.

The first phase seems to be Ethereum's launch (prevalent in topics 29 and 4), the DAO hack and split in 2016 and the initial post-split incremental change in PoW via the Istanbul fork (prevalent in topics 29 & 20). The second phase is from the incremental Istanbul fork up to the Beacon fork (prevalent in topics 29 and 20). That fork is aimed at making a more competitive and secure PoW system, and also to introduce elements of PoS into Ethereum's smart contracting approach (prevalent in topic 20). The second phase therefore challenges the original democratic imaginary (prevalent in topic 29) and begins to focus on competitive positioning and creating new, more scalable forms of smart contracts (topic 4). The third phase, like the second, starts with the incremental fork, the London in 2021, and runs through the big Merge fork of Sept. 2022. The Merge is the complete switchover to PoS and its aftermath. Not surprisingly, it involves all three focal topics (prevalent in topics 29, 20, and 4).

4.1. Phase 1 – Pressurized possibilities

Ethereum, first proposed in 2013 by Vitalik Buterin, was implemented in 2015 by a small group of developers branching off from the Bitcoin community. These development efforts were initially guided by a document authored by Buterin called the “Ethereum Whitepaper” (prevalent in Topics 29 and 4). At the time, Buterin had been writing for a Bitcoin-based magazine and advocating for a way for Bitcoin to facilitate application development on the platform (in what was known as a “scripting language”). His initial efforts did not gain traction with the Bitcoin audience; as a result, he proposed a new platform. In the white paper, he explicitly contrasted Ethereum with Bitcoin:

“Bitcoin in 2009 has often been hailed as a radical development in money and currency, ... However, another, arguably more important, part of the Bitcoin experiment is the underlying blockchain technology as a tool of distributed consensus ... What Ethereum intends to provide is a blockchain with a built-in fully fledged Turing-complete programming language that can be used to create “contracts”, allowing users to create any of the systems described above.” (Buterin, 2014: 1).

The platform was articulated in technical detail, including the technical architecture and specific aspects of implementation. It also articulated a philosophy motivating this and outlined possibilities for both applications and use cases. Like other blockchain platforms, Ethereum at the time adopted the Proof of Work (PoW) architecture. Although there was much shared with Bitcoin, the contrast were clear. A prominent theme of the document was applications build with the advanced scripting language. In this theme was a push to move beyond only offering an economic layer (cryptocurrency) and to enable decentralized applications (the foundational layer). As Buterin wrote, “there is also a substantial array of applications that have nothing to do with money at all.” (Buterin, 2014: 34). Throughout the document, democratic ideals were discussed alongside the concept of decentralization, including an explicit discussion about “online voting and decentralized governance” as a core use case.

It was initially materialized and built out by Buterin and a small group of software developers. They translated the white paper into a roadmap artifact that established a linear path of stages, each with names and target dates, “Frontier” (2015), “Homestead” (2016), “Metropolis” (2017), and “Serenity” the “final destination”. The platform was initially implemented as an open-source ecosystem that was “permissionless, decentralized, [and] censorship resistant” (Buterin, 2023) that drew upon wider, “cypherpunk” values (Swartz, 2018). To combine these local field elements with the new community principles, Buterin and other entrepreneurs (Hoskinson, Di Iori, Alisie, Chetrit – and a bit later, Lubin, Wood, and Wilcke) founded a non-profit organization in 2014, Zug, Switzerland called The Ethereum Foundation. They subsequently organized a first conference for developers in Berlin in November in 2014 called “DEVCON-0”, followed by DEVCON-1 in 2015. A community logic was enacted by the early Ethereum community – one that attracted developers interested in smart contracting, along with purposeful leadership.

4.1.1. The hack as a shock event

Amidst the rapid growth, in 2016, the community faced an existential crisis. A prominent hacking incident occurred just as the Ethereum community was attempting to set up the first decentralized autonomous organization (DAO) on the platform using smart contracts (Hsieh, Vergne, Anderson, Lakhani, & Reitzig, 2018). This was envisioned in the initial decentralized technology imaginary (Dylan-Ennis et al., 2023) proposed by Buterin in the “whitepaper”, but it had not yet been implemented in practice. In this first attempt at implementing a DAO, the community was split over *how* to do it and the underlying justification. A substantial amount of currency (\$60 million of Ether in 2016 USD) was stolen – ostensibly by a disaffected member of the community leveraging a publicized security hole. They purportedly wrote an open letter to the community about it as a form of ideological protest against what they saw as moves by Buterin and the Ethereum Foundation that were violations of the “code is law” value in crypto (DuPont, 2017). However, the response to this *shock* is what really challenged the Ethereum community's original decentralized imaginary:

“Moderates’ saw the hard fork as evidence of the flexibility and practicality of Ethereum and its leaders, while the more ideological saw the hard fork as censorship by a powerful cabal, or proof that blockchain technology was unable to live up to its idealistic promises.” (DuPont, 2017: 165).

In the response to the hack, the Ethereum community deliberated for almost a month and proposed to intervene and modify the blockchain code – a contradiction to the community's original governance principles (Lehdonvirta & Robleh, 2016). It also proposed to increase auditing and, via the forking, essentially preserve the lost currency from the hack. The changes were approved through a series of votes for collective decision-making, resulting in a governance model where the Ethereum Foundation and developers lead, with substantial backing from miners. The democratic decision resulted in a hard fork, a permanent modification to Ethereum's technology architecture that diverged from the existing blocks. This led to a split in the community and the creation of a forked blockchain platform: Ethereum Classic (ETC) versus the Ethereum platform (Hannigan et al., 2022). The latter was closely associated with the Ether cryptocurrency, which grounded it further in the wider blockchain community (Dodd, 2018).

4.1.2. Technology and governance responses

The DAO hack shock, in a somewhat *reactive* way, ushered in a shift of technology and governance for the Ethereum community. It became clear that Ethereum technology was promising for applications well beyond the Ether cryptocurrency, but security needed to be addressed. Efforts to shore up security concerns were taken seriously by the community, particularly as vulnerabilities in the smart contracting infrastructure led to exchanges such as QuadrigaCX and Parity losing significant amounts of money (Zachariadis et al., 2019). Because many of these actors were organizations dedicating people to contributing to the open source Ethereum blockchain source code, they had a direct interest in addressing security concerns. As a result of work to fix security vulnerabilities, by 2017, dApp usage on the platform grew significantly (Wu, Ma, Huang, & Liu, 2021).

The norms and roles of the community were expanded and codified. These included roles and incentives for miners to generate blocks (and track transactions) and developers to build the technology architecture (Table 1; Supp. Tables A1 & A4). The policies of the community (i.e. rules of the game), through communications, conferences and articulation became better understood by community members, grounding the group ontologically (Giddens, 1991). The policies as implemented in the technology architecture were referred to colloquially by community members as “the protocols”. When the platform was initially proposed, it was pitched through a set of idealized possibilities around decentralization. As the platform materialized and the community was built, these possibilities were pressurized.

4.2. Phase 2 – Innovating and garnering resources

The DAO Hack and fork, along with technical developments in blockchain modeling, allowed for greater scaling with dApps being hosted on the main Ethereum platform (Zachariadis et al., 2019). However, Ethereum became increasingly criticized for its lack of ability to handle the platform's capacity as work and the processing of transactions grew; i.e., in terms of scalability. For example, when the game CryptoKitties took off as a viral phenomenon, this crowded out other applications and led to both congestion and increased transaction costs for mining on Ethereum (Serada, Sihvonen, & Harviainen, 2021). The issues around scalability increased substantially as more dApps were created on the platform. With increased use, the transaction fee developers paid to the miners (i.e. the “gas” fee) increased, which led to significant pressure on core members of the community to act (Dahlander & Frederiksen, 2012).

4.2.1. Distributed experimentation

Ethereum began *experimenting* with a new organizing vision (Swanson & Ramiller, 1997) for maintaining the platform called “Proof-of-Stake”. The concept was pioneered in a limited manner on the crypto platform Blackcoin in 2014 (Hsieh & Vergne, 2023). For Ethereum, the proof of stake concept initially appeared to be a more robust approach to scaling and security issues, and would position the platform better for future regulation:

“Let's look at attack resistance. ... it pushes strongly in favor of proof of stake over proof of work, as computer hardware is easy to detect, regulate, or attack, whereas coins can be much more easily hidden (proof of stake also has strong attack resistance for other reasons)”

(Buterin, 2017)

The process of imagining a future that became known in the community as “Ethereum 2.0” around the transitioning to PoS was a

gradual one, a “distant imaginary” (Augustine et al., 2019). A long-term release plan document was created that consisted of various developmental stages including an initial transition to PoS (Gupta, 2015). This was followed by a series of collaborative democratic community efforts. Through developer conferences (i.e. Devcon) and online forums like Reddit and Twitter, Vitalik Buterin and the Ethereum Foundation, developers, miners, and the broader community actively engaged in discussions about Ethereum's technological advancements, challenges, and future directions. Buterin took the important step of assembling many of the ideas being floated in the community and created a road-map document entitled “A Modest Proposal for Ethereum 2.0” (Devcon 3, 2016). It outlined some of the principles of a Proof of Stake architecture (PoS), including staking, new hash-tagging, and a method of audit. When presented at the Devcon 3 conference in 2016, this talk served as a catalyst for further discussions and development, outlining initial ideas for a novel practice called sharding and for the PoS implementation.

As part of this gradual development, there was a need for more of a *distributed* version of their experiments in order to handle the difficult community-level coding changes required. Broad architectural shifts such as the move to PoS were instantiated as multiple experimental (and often overlapping) EIP proposals, some in a state of partial materialization:

“...The two core components of the phased [Ethereum] 2.0 transition are incredibly difficult to engineer for a decentralized system of this breadth: a foundational shift from a proof-of-work consensus algorithm to proof-of-stake blockchain, and an ambitious ‘sharding’ effort to dramatically accelerate transaction processing. ... With no corporate organizational structure to direct commands, all work is conducted by a giant, decentralized, hard-to-coordinate team, with armies of developers in far-flung parts of the globe earning bounties to **detect bugs in the system's sprawling codebase. That's difficult...**”

(CoinDesk, 2020, June 19)

There was also a need for financial services and investors to work with the new code as part of that experimentation (CoinDesk, 2020, June 19). It did so by creating a PoS-based test platform (a “testnet”) that they named “Beacon chain”. This operated on a separate platform (blockchain and architecture) from the main Ethereum platform (known as the “mainnet”).

4.2.2. Staking as anteing-in

In the initial framing of this test platform, it was proposed that the Beacon chain would depend on a new role of third-party validators. This entailed creating a registry of validators who stored a certain amount of resources on the platform. Maintaining the public ledger would then be based on each block (or set of contracts being validated) being processed by validators chosen at random from the registry (Supp. Table A1). Staking was the label assigned to this practice. It was introduced in the Ethereum community as an analog role to mining but with a twist (Ethereum Foundation, 2024b). In this practice, users would put up Ether as a “stake” (similar to the concept of collateral), which would enable them to activate software that would create and validate new blocks on the Ethereum blockchain.

Preparations for the Beacon chain included establishing a minimum deposit of 32 Ether as a stake for users to participate. In this implementation of the PoS architecture, the practice of staking was central, and this meant that users would ante-in (i.e. invest) resources to participate. To stake meant locking up resources (Ether) for a set period of time until the promised Ethereum 2.0 was released, but for users willing to participate in the Beacon chain test, this also meant a guaranteed return (listed as an annual percentage rate). In addition, the creation of the PoS testnet needed to fundraise a minimum of 524,288 Ether from validators.

It was very difficult to persuade miners to accept a new protocol and commit their own resources. Many were worried that the new platform payment per transaction – “gas fees” might reduce the miners' earnings (Cointelegraph, 2020, June 13). To signal his commitment to Ethereum 2.0 and deal with this resistance, Buterin staked as well:

“Ethereum founder Vitalik Buterin has sent his first ether for staking on the next iteration of the blockchain network. The transactions were sent to Ethereum 2.0's newly launched deposit contract, which went live Wednesday as a means for network participants to move funds from the current proof-of-work blockchain to the soon-to-launch proof-of-stake (PoS) blockchain....”

(CoinDesk, 2020, November 6)”

4.2.3. Security issues and anticipating regulation

The security threats to the existing Ethereum mainnet platform (based on PoW) were still being debated in 2020, particularly after the Ethereum Classic (ETC) branch (i.e. those still using the old fork after the DAO hack) was breached by what was known as a “51% attack”. Here, when a group temporarily took over more than 51% of the mining power, they could manipulate blocks on the chain. This was a well-known security vulnerability inherent in the PoW architecture. In August 2020 there were three separate 51% attacks to ETC (Cointelegraph, 2020, September 3). This resulted in the disruption of over 10,000 blocks and millions of dollars in losses.

Although the Ethereum Foundation primarily dealt with the Ethereum mainnet platform, they still maintained some ties to the ETC platform. Rather than pushing for dealing with this issue quietly to avoid internal revolts, external publicity, and shared reputational hazards, the Ethereum Foundation responded openly and proactively by offering bounties for testing and identifying bugs to potential developers, underscoring the urgent need to enhance security measures on the Ethereum main platform (and its related forks). The engagement of miners was crucial because, as publicized in *The Cointelegraph*:

“A potential protection against chain reorganization [like the recent attacks to ETC] is setting up checkpoints at a node level which would set the history of the blockchain in stone after that point. Any proposed blockchain changes beyond this checkpoint would thus be rejected by nodes.” (The Cointelegraph, 2020, August 7)

After these hacks on Ethereum Classic (using the PoW-based architecture), Vitalik Buterin commented: “The ETC should just switch to Proof-of-Stake. Even given its risk-averse culture, at this point making the jump seems lower-risk than not making it” (*Cointelegraph*, 2020, August 7). Some competitors returned to debate the issue. Charles Hoskinson, the founder of Cardano and one of the original five co-founders of Ethereum, said he would help “...the troubled Ethereum Classic community ... with a major condition [that] the community would first have to institute a decentralized treasury system, similar to Hoskinson's Cardano [a separate crypto platform] and many other blockchain projects.” Terry Culver, a leader of ETC developers, rebuffed Hoskinson for his suggestion: “As an open-source, decentralized community...we're not waiting for someone else to step in” (*Cointelegraph*, 2020, August 10). At this time there was significant tension in the broader Ethereum community.

The point about “waiting to step in” touched on a related issue: *concern about being regulated from outside*. Technology that seemed more secure could preserve Ethereum in ways to avoid such external meddling. Terry Culver in September of 2020, said: “It's not a question of abandoning proof of work. It's a vulnerability that all proof-of-work blockchains have, even Bitcoin and Ethereum” (*Forkast News*, 2020, September 7). His preferred adjustment raised governance as the solution: “...the industry as a whole wants to be able to self-regulate and have an environment where the government doesn't need to be in the weeds.” One point for persuasion was to offer Beacon and similar PoS tests as a means to move toward an ultimately “safer” and more secure platform. While these debates dragged on, Ethereum's main platform continued to suffer from ongoing security vulnerabilities (Rameder, di Angelo, & Salzer, 2022).

In December 2020, the Ethereum foundation finally gathered a sufficient validator pool- made up of users and their Ether resources – to launch its own new test platform under the aforementioned name “Beacon chain”. The introduction of Beacon chain marked a pivotal shift in Ethereum's architecture from a PoW-based mainnet to a dual parallel system that included the PoS testnet. Though the change did not impact end-users' experience, this transition altered the institutional landscape by the co-opting validators pool into the Ethereum community. During this time, Ethereum was able to attract various decentralized applications like DeFi, NFTs, stable coins, and DAOs to the mainnet. Ironically, Ethereum miners were initially the most satisfied. As the transaction rapidly increased due to the surge of dApps built in Ethereum, the miners could exploit profits from the large number of transactions. In May 2021 gas fee earnings rose to their highest peak in the platform's history (*Cointelegraph*, 2021, June 9).

4.3. Phase 3 – Transforming the tech-mediated community & its positioning

After the successful launch of the Beacon chain as a testnet platform in December 2020, Ethereum prepared for the London Fork to deal with some of the ongoing issues around scalability (congestion and high transaction fees) on the mainnet platform. Scheduled for August 2021, it was again anticipated that initially miners' profits would be significantly reduced, but it was a necessary stepping stone during the broader transition to PoS. In a sense, the prior path of development was being followed by Ethereum in a methodical fashion. By committing to this path and addressing prior issues around scalability, security, and decentralization, it gained further legitimacy: it continuously attracted dApps from DeFi to NFTs. This influx of dApps brought “monumental gains in value and the increased volume of transactions and users” (*Cointelegraph*, 2021, February 24).

However, the changing composition of the field and the scaling up of crypto currencies also brought new regulatory challenges - ones that could result in external, uncontrolled shocks to the Ethereum community if not handled proactively. A key issue was sustainability. As the price of cryptocurrencies continued to rise as well as the widespread diffusion of dApps during COVID-19, the energy consumption and its environmental consequences associated with PoW mining rapidly increased, attracting further regulatory attention from governments. Many countries like China worried tremendously about the amount of energy that blockchain mining was taking. After all, China accounted for “65 percent of the total global electricity use from crypto mining, with the northwestern Xinjiang Province, where coal is abundant, serving as its heartland” (*The Independent*, 2021, June 18). Furthermore, nation state governments were grappling with the legal ambiguity surrounding these decentralized applications. China had serious concerns about capital outflows through unclassified applications. In September 2021, Chinese authorities banned cryptocurrency trading and mining across the board.

4.3.1. Emotional issue framing

Several blockchain fields and crypto platforms felt the effect. The two largest PoW-based blockchains' – Bitcoin and Ethereum – cryptocurrency values tumbled. A prominent art portfolio platform was “bullied into not selling NFTs less than 24 hours after announcing plans to launch NFT artwork from multiple notable artists” (*Cointelegraph*, March 10, 2021). As one expert pointed out, “instead of constantly traveling the world hunting for and monetizing bad grid design and fossil fuel waste”, blockchain communities, including Bitcoin miners, must “behave like every other energy-intensive industry and participate in the complicated social, political, and economic process of finding a place in a zero-carbon, energy-constrained world” (*The Independent*, 2021, June 18). This was a challenge, in other words, to the underlying financial market logic for the crypto field itself.

Bitcoin advocates defended PoW and tried to avoid admitting the issue, saying “mining is increasingly being done with electricity from renewable sources as that type of energy becomes cheaper, and the energy used is far lower than ... the energy wasted by plugged-in but inactive home devices in the US” (*The Guardian*, 2021, February 27). In contrast, Ethereum did not jump to the defense of PoW. Senior members of the Ethereum community viewed blockchain validation and sustainability as compatible. Buterin employed an *overtly emotional framing* to put forth this case. In an interview with *Time Magazine*, he “explained the ‘dystopian potential’ of digital assets if implemented incorrectly” (*Time*, 2022):

“... he fears that his vision of creating a more egalitarian digital economy risks being overtaken by nefarious actors who are only after greed. ‘If we don't exercise our voice, the only things that get built are the things that are immediately profitable,’ he said ...

The interview painted Buterin as a pragmatic leader taking a ‘middle ground’ approach to solving issues that impact the community... While Ethereum's evolution to a proof-of-stake chain remains mired in delays, the investing community has high hopes for the future.”

(Cointelegraph, 2022, March 18)

Also notable during this period was a shift in framing to match Ethereum's expanding audience. With increased attention on dApps, Ethereum's discourse began to reflect concerns from state regulators. Buterin leaned further into the link between PoS and sustainability:

“In an interview with Bloomberg, Buterin said the [London] fork will help lower its carbon emissions by 99%. The London upgrade is “proof that the Ethereum ecosystem is able to make significant changes.” ”

(Business Insider, 2021, August 7)

In this interview he directly addressed how the Merge fork (i.e. the “London” fork) would lower carbon emissions. But he also worked to maintain legitimacy of this change by pointing to the London fork as evidence that the community was able to undergo significant changes.

4.3.2. Building a new vocabulary and philosophy

In the summer of 2021, Ethereum was busy preparing the London fork for the mainnet platform. The fork's aim was to introduce a base transaction fee and protect the blockchain from a 51% attack by burning the base fee. This deflationary process-and-switch was expected to reduce (temporarily) miner's profits by 50%. The Ethereum Foundation was compelled, therefore, to cooperate with miners to ensure a secure forking process, while trying to exhort them to reduce their short-term interests just after the forking event. At the same time, Ethereum was threatened by competing platforms, both for mining under PoW, and for PoS validation (e.g., Solana and Cardano) – platforms to which miners could defect. An expert pointed out, “Ethereum still has by far the largest developer community, as well as the number of dApps, built on it, but still, complacency is a killer” (Cointelegraph, 2021, March 21).

Rather than taking a two-pronged approach, such as had been done with forking that created Ethereum Classic and Ethereum, Buterin and the Ethereum Foundation, emphasized at the Scaling Ethereum Summit conference in April 2021 the danger of doing so and proposed a new method of architecture to match the Ethereum 2.0 vision: layering one architecture over the other, with staggered timing, then merging them. “We don't want to do all the potentially dangerous things at the exact same time. You want to do the first one [Ethereum's Merge to PoS] and then the other so that developers can pay attention” (Coindesk., 2021, April 28). The mainnet platform and testnet platform to be created would be layers within the same system, gradually combined and further legitimating PoS:

“Danny Ryan [a key Foundation researcher] explored how we could accomplish this by leveraging the existing Eth1 clients in his Eth1+Eth2 client relationship post. This would massively reduce the development work required to deliver a post-merge system and leverage existing clients, which had been battle-tested for years on Mainnet. Around the same time, research on rollups as a viable and secure way to scale Ethereum proved promising. Instead of waiting on a complex, uncertain scaling solution years away, we could shift the focus towards scaling via rollups instead of sharded execution.” (Ethereum Foundation, 2022).

As can be seen, this architectural finesse required a new set of terms and ways for the Ethereum technology community to grasp the changes. As the Ethereum foundation wrote, “As work began on the Beacon Chain, it became clear that the phased Ethereum 2.0 roadmap would take several years to deliver fully” (Ethereum Foundation, 2022). The Foundation recognized that the community needed to shift “mental models”, deal with “inclusivity”, “scam prevention”, and “staking clarity”. In the new vocabulary these Eth1 and Eth2 terms (short-hands for Ethereum mainnet and Ethereum 2.0) were relabeled and defined:

“Beyond protocol development, a critical shift in Ethereum has been the movement away from ‘Eth1’ and ‘Eth’ terminology. As of late 2021, core developers stopped using the terminology, preferring “execution layer” and “consensus layer”, respectively. Today, as highlighted in our Q1 roadmap, ethereum.org makes the same shift.”

(Ethereum Foundation, 2022)

The new platform and technology *community vocabulary* reinforced the changes being advocated by Buterin and the Foundation. By reframing the mainnet platform and testnet platform as layers within the same interconnected system, this further legitimated PoS. Updating the roadmap set the stage for a merging of the two platforms into one.

In a further effort to legitimate the staking practice in PoS, Vitalik Buterin articulated a *new philosophy* whereby the community was seen as a “network state”. On his blog, he began embracing a visionary idea by a serial entrepreneur named Balaji Srinivasan. In his 2022 book “The Network State: How To Start a New Country”, Srinivasan challenged the basic system of nation-states and proposed an imaginary of sovereign states attached to online communities that crowdfunded territories. He described “a network state is a highly aligned online community with a capacity for collective action that crowdfunds territory around the world and eventually gains diplomatic recognition from pre-existing states”. (Srinivasan, 2022: 9). Although Buterin initially critiqued the centrality of elites (“recognized founders”) and the political angle of Srinivasan's idea (the concept of “megapolitics”), he embraced the network state concept,

“In my view, the problem of how to settle into something other than founder control is important, and Balaji spends too little effort on it. “Recognized founder” is baked into the definition of what a Balajian network state is, but a roadmap toward wider participation in governance is not. It should be... On July 4, Balaji Srinivasan released the first version of his long-awaited new

book describing his vision for "network states": communities organized around a particular vision of how to run their own society that start off as online clubs, but then build up more and more of a presence over time and eventually become large enough to seek political autonomy or even diplomatic recognition. Network states can be viewed as an attempt at an ideological successor to libertarianism"

(Buterin, 2022)

Buterin connected the network state concept with the community logic that was prominent in the Ethereum community:

"Network states, with some modifications that push for more democratic governance and positive relationships with the communities that surround them, plus some other way to help everyone else? That is a vision that I can get behind". (Buterin, 2022).

What is important here is the way that Buterin begins to shift the philosophy. In his post, he uses Srinivasan's book (as well as David de Ugarte's book) to explore the importance of collectives. The framing shifts away from decentralization as a key value (carried forward from the cypherpunk era) and leans into the value of collectives. This shift in framing matches the shift from PoW to PoS.

4.3.3. *Creating additional roles and interdependencies*

Despite the successful Beacon test and enrolling miners into the experiments – plus fending off challenges from regulators around sustainability and security – the situation for Ethereum worsened by the end of the summer of 2022. Miners and their cryptocurrency values were particularly hard hit. After the Beacon and London forks, most of the tensions and pressures in the community had come from the developers who were suffering the consequences of congestion on the platform in the form of higher transaction fees and slower performance.

The shift to a two-layer system of the mainnet platform and Beacon chain testnet platform and reframing with new vocabulary was sped up some with the notion of an "early merge" (Ethereum Foundation, 2022). In this, there was a more explicit division of labour between the two layers. The mainnet was given further runway as a PoW system called "the execution layer". What was formerly known as the "Beacon Chain" testnet platform was rebranded as the "consensus layer". As a key part of this shift Ethereum Foundation and core developers tried to reframe *roles and interdependencies*. The PoS-based system was to be operated by validators who had staked Ether; whereas miners could serve as validators in the "consensus layer". This change reflected a shifted division of labor, whereby the validators would initially take over the duties of running the block-based elements in the technology infrastructure – including duties around block proposals, attestations, and sync committees. The miners' duties would initially be limited to processing transactions, smart contracts, and the Ethereum Virtual Machine (EVM) as part of the "execution layer". Put differently, it was also framed as the "battled-tested" elements of transaction processing being initially run by PoW miners, while PoS validators would gradually take over other infrastructure duties (Ethereum Foundation, 2022). Of course, the new arrangement of duties meant that, in spite of the "early merge" terminology, the eventual transition to a single system might take longer.

As part of this use of language and redefinition of roles, Ethereum commentators identified alternative options, naysayers, and defectors in discussions. And the commentators did so using somewhat villainous terminology. Community forking efforts, such as Ethereum Classic, were relabeled as "zombie" networks, clearly not a positive sentiment. Blockchain media commentator *CoinDesk* suggested that such networks and their defector members were becoming irrelevant and populated by those members the new Ethereum did not want:

"I think there are two candidates that could play spoiler in 2022. The first is a film classic: zombies. Blockchain Zombies are layer 1 networks that are sitting on piles of cash and haven't accepted defeat at the hands of Ethereum. They are the Ethereum killers that couldn't. They're not ready to go quietly into the night and they will fight irrelevance with big cash outlays to developers. I won't be surprised at all to find some of them newly re-branding themselves as layer 2 networks under the if-you-can't-beat-'em-join-'em rule. It's already happening, but expect these activities to take on a new level of urgency in the coming year."

(CoinDesk, 2022, January 5)

Thus, there was a significant effort to build bridges in the Ethereum community with the reframing of separate platforms (Eth1 and Eth2) into a single platform with two layers (consensus layer and execution layer). What began as the "early merge" was then formalized into proposals for a fork which would be called "The Merge" (Cointelegraph, 2022, January 4). At the end of the day, Ethereum, technologically, had little choice but to adopt the Merge. Still, the community members needed to hold its collective breath and hope that in the September of 2022 the group would remain intact, supportive, and vibrant. Some miners did defect, and mining disappeared from the platform. Validators and staking arose in their place. At the same time, Ethereum's energy use also dropped by 99%. The sustainability frame allowed regulators to distinguish Ethereum from Bitcoin cognitively, focusing on the differences in their new foundational technologies. This distinction was evident in the White House's report published in September 2022 (Wired, 2022, February 9), right before the Merge. The report expressed strong concerns about Bitcoin's PoW's substantial energy consumption and considered PoS, used by Ethereum's transition, as a plausible alternative.

5. Discussion

This paper was animated by the puzzle of how a decentralized tech-mediated community might be able to scale yet retain its vital character (Faik et al., 2020; Massa, 2017; O'Mahony & Bechky, 2008). Ethereum as a type of tech-mediated community was a modal

alternative to Bitcoin and other crypto currency communities. Early Ethereum members, particularly miners, engaged with PoW protocols and held to the cypherpunk values and anarchic aims, which fit a decentralized democratic community logic (DuPont, 2019; Dylan-Ennis, 2024). This technology, its users, and the underlying logics were somewhat entrenched as a socio-technical assemblage (Caliskan, 2020; Hayes, 2019), so it was not at all clear how to transition to a new type of blockchain protocol with new roles – i.e., to PoS – and a different underlying logic that prized audit and self-regulation along with the notion of a democratic community (Buterin, 2023). But only by solving this problem could the scaling of Ethereum be achieved.

5.1. A process model of prefigurative enrolling

Our co-constitutive approach and findings showed that processes around prefigurative enrolling offered at least one credible way to resolve the scaling problem. In the model, there were three successive phases based on setting up and getting support for the various technology changes, punctuated by key hard forks, all sitting atop field logics (see Fig. 3). The first phase was *pressured possibilities*. In it, organizational possibilities (Hannigan, Briggs, Valadao, Seidel, & Jennings, 2021; Lounsbury & Glynn, 2019) were initially theorized in the “whitepaper” but then required rejigging and rediscovering in the face of competitive pressures and *shock events*. These, in turn, led to *reactive modifications* in the technology and governance system and a return to the original roadmap, such as we saw with the splitting fork Ethereum Classic versus Ethereum (Pak, Hannigan, & Jennings, 2023). At this point, the community still reflected and drew upon that earlier group’s primarily (democratic) community logic (Hedberg & Lounsbury, 2020).

In the second phase, on of *innovating and garnering resources*, we saw more of a typical mobilization story (Davis & McAdam, 2000). Enrolling members of a tech-mediated community requires renewed innovation (and experimentation) to maintain the attraction of members to the community and retain its original group (Grodal, 2018; Massa, 2017). In order to scale, members needed to commit resources – time and money – more formally to this process. Cleverly (though some might think riskily), the community founders requested commodity and time-based *stakes* into the community. Yet this innovation and resourcing process was constrained by ongoing events around *security and regulation*, which required internal, planned for, modifications. As the field around the alternative platform expanded, even smaller or alternative platforms were making money and becoming visible, pulling in more of the finance logic from outside in.

The third phase of the prefigurative enrolling process is a *transformation of the community and its position*. There was a widening of the community boundary and zeitgeist that hooked onto broader field issues, such as sustainability, and a *framing of the issue(s)* in a way that is relevant to the technology community. We found this framing to be emotive, yet also related to the platform technology skills, such as coding. New possibilities were identified and elevated to become elements in a new roadmap for the community. The new imaginaries relied on *new language*, such as around PoS, and a modified *philosophy*, such as about the role and meaning of validation (versus mining). A complementary part of this widening process is to *adding new roles* and clarification of less acceptable roles and practices – the zombie platform members and PoS naysayers. The new roles (and anti-roles) required additional coordination mechanisms, which increased interdependencies (Hsieh & Vergne, 2023). The framing of key issues, like sustainability, in acceptable terms for the community and sharpening the language and philosophy around it reinforced the sense of new opportunities for members, and, at the same time helped *blend* competing market logic features (e.g., around cryptocurrency) with those from the community logic. Phase three ended, therefore, with a consonant, co-constitutive shift in technology, the technology community, and its underpinning logic.

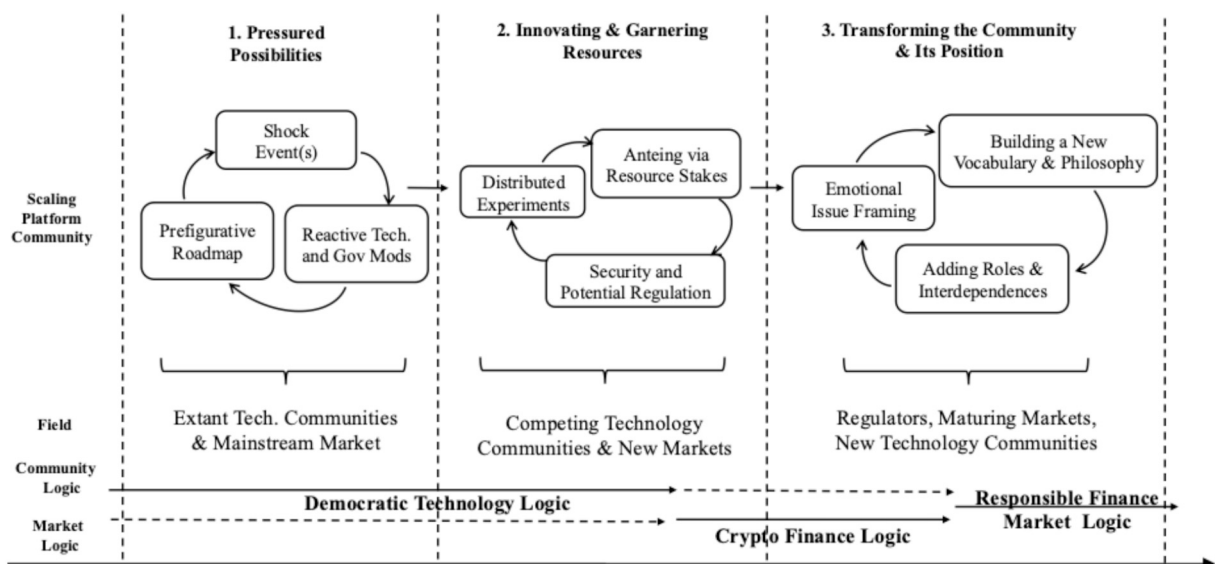


Fig. 3. Ethereum's prefigurative enrolling mechanisms.

5.2. Contributions

Our paper's primary contribution is to the literature on tech-mediated community scaling. As shown in the theory section, there have been credible answers to addressing the scaling issue, including diffusion of technology forms and practices to fertile fields that adopt and adapt them (Audia et al., 2006; Barnett, 1990), and, in contrast, the creation of tech-mediated communities that then scale (Orlikowski & Barley, 2001; Saxenian, 1994) and link to other communities that transport in their talent and techniques (Powell et al., 2005). More co-constitutive approaches (Faik et al., 2020; Hedberg & Lounsbury, 2020; Lounsbury & Wang, 2020) argue that these macro and more micro field processes are combined as a tech-mediated community scales based on its platform and core techniques. However, with so many potential theoretical components to detail and investigate empirically in such models, we still do not know enough about the mechanisms involved in the early scaling process in nascent fields nor how logics are leveraged and enacted through a tech-mediated community's scaling mechanisms – and with what consequences for community logics. Although a nascent field is not a boundary condition for our theorizing, this setting is well suited for observing cultural entrepreneurship amidst scaling efforts.

Our prefigurative enrolling model and the longitudinal case suggest, for Phase 1, that road mapped imaginaries by core groups, ones tied closely to the technology features and affordances, are critical early on. This is more similar to Santos and Eisenhardt's (2009) early narratives and market claims of Silicon Valley technology firms in nascent markets rather than to the importance of multiple projects mediated by the platform (O'Mahony & Bechky, 2008). These form a decoupled alternative community (Howard-Grenville et al., 2017) or demonstrate distinctive professional competence as a group (Croidieu & Kim, 2018). In addition, in our enrolling model, the technology community faces contestation and tensions as key early moments (Howard-Grenville et al., 2017; O'Mahony & Bechky, 2008). However, we found these shocks to be quite sharp, threatening the imaginary and fabric of the community. The possibilities of the group were (Lounsbury & Glynn, 2019), in other words, highly pressurized. More reactive and immediate responses in technology and governance terms were needed early on, followed by a return to the roadmap – and round – until the community was prepared for more entrepreneurial scaling (Phase 2).

The core of the model (Phase 2) appears, on the face of it, to be similar to scaling in opportunity exploitation (Shane & Venkataraman, 2001) or mobilizing new technologies (Grodal, 2018; Howard-Grenville et al., 2017). That is, there is an innovation to which multiple new players (often in a network) are committed, resources from that group are anted-in and the group's boundary is enlarged to include nearby groups and their resources (time, money, support). In addition, surrounding field issues relevant to technology communities must be considered during this exploitation phase, such as security (e.g., IP), regulation, and competition. Yet in our longitudinal case of enrolling, each of these more routine scaling mechanisms takes on a different flavor. Experimentation occurred in a distributed fashion (Garud & Karnøe, 2003; Lounsbury & Glynn, 2019) and around more radical variants. It is also standard organization experiments that take place in loosely coupled (Weick, 1976) subunits or “skunk works” (Peters, 1983), or depend on isolated “fail-fast” trials (Ries, 2011). There was also anting-in in via stakes around the technology itself, hence not just some occupational participation beyond membership fees, training, and new practices, but investment in the technology by skilled workers, such as one might see with cooperatives (Hedberg & Lounsbury, 2020; Rothschild-Whitt, 1979). Furthermore, because there may (or may not) be multiple rounds of cycling through the mechanisms in Fig. 3. The model accords with either rapid scale up (Santos & Eisenhardt, 2009) or relatively slower (Gawer & Phillips, 2013; Hargadon & Douglas, 2001).

The Phase 3 of prefigurative enrolling, unlike classic discursive mobilization models (Benford & Snow, 2000), is where substantial emotive framing and engagement with the external macro institutional environment occurs. This use of framing at this stage is different from the ordering in Faik et al. (2020), which has sensegiving, then translation, then decoupling, and perhaps more akin to Hedberg and Lounsbury (2020), where recoupling occurs after the period of development in phase 2. The enrolment model shows the initial important parts of recoupling as a revitalized philosophy and new language, then adding in more roles and responsibilities that are attuned to these cultural features. These mechanisms also help build up the occupational community (Croidieu & Kim, 2018; Howard-Grenville et al., 2017). Finally, we argued that via discussion of the roadmap and emotional framing of issues, like sustainability, the community's imaginary and logic(s) can shift in more evident ways. The shifts we documented allowed for at least two logics during any phase's configuration (Greenwood et al., 2011), plus the possibility of blending them at the end into something new along the way (Pache & Santos, 2013).

The set of prefigurative enrolling processes captured in Fig. 3 contributes to the literature on prefiguration. Prefiguration was first developed in sociology with social movements as a way to show how activists work to create futures in the here and now through ongoing experimentation (Graeber, 2002; Yates, 2015). Organizational studies extended this with a focus on exploring provisional futures, telescoping them to the present, and finding accommodations with institutional environments (Reinecke, 2018). A separate strand of literature in Science and Technology Studies (STS) build on the concept of “sociotechnical imaginaries” (Jasanoff & Kim, 2015) to show how technological affordances are used to prefigure alternative futures (Dylan-Ennis et al., 2023; Swartz, 2018). By bringing together the STS material into organizational studies, we show how sociomateriality is important in prefiguration, particularly with tech-mediated communities. We also bring in cultural entrepreneurship and its focus on institutional logics, to show how prefiguration processes need to enroll collective support and scale. In this case, it is anchored materially in the tech-mediated affordances of blockchain for its community.

Our model of prefigurative enrolling contributes to the cultural entrepreneurship (Lounsbury & Glynn, 2019) in several ways. We take its premises of institutions, actors, tools, imaginaries, and material circumstances seriously, but then shine a light into the processes of scaling without downplaying its complexities. In the Ethereum case, the blended-logic embraced elements of consensus and sustainability, but also with novel financial goods (cryptocurrency). As a result, Ethereum was able to scale not only in terms of membership, normalized asset base, and reputational reach, but as an alternative form of cryptocurrency-creating community. All too often, when authors use the conceptual term “cultural entrepreneurship”, they presume actors and future states are already constituted

and that individually are agentially able to mobilize others in the field. Our work confirms that cultural entrepreneurship that is culturally embedded, but shows this is not a simple process.

5.2.1. Methodological contribution

A secondary area of contribution is methodological in nature. Technology community dynamics have been examined using discursive data (Bechky, 2003; Leonardi & Barley, 2010; Orlikowski & Barley, 2001). The actor interests and frames (Grodal, 2018; Gurses & Ozcan, 2015; Reinecke, 2018) have been described in detail with them, capturing actor mobilization (Benford & Snow, 2000). At the same time, broader level field dynamics have been captured using larger corpora around many actors, with clustering or dimension reducing methods to determine cultures and logics (Croidieu & Kim, 2018; Meyer & Höllerer, 2010). Consistent with our co-constitutive approach, we have relied on this more localized and field level approach using very micro data from field blogs, communiques, and commentaries close or within the community (Massa, 2017), but also wider sets of commentaries and issues in the media around such issues (Grodal, 2018). We have rendered a textual corpus (Hannigan et al., 2019) the sits together to pull out their joint threads, then using structural topic modeling as a somewhat newer technique (Villamor Martin et al., 2023), tied these to specific actors and time points or issues. Using these *focal topics*, we were able to examine the co-constituted topics, such as the prefigurative roadmap, without dividing the data and interpretation along micro versus macro lines.

5.3. Limitations and future research

Longitudinal single case studies such as Ethereum's scale up, always face a key limitation – generalizability. Being of a single case and limited set of periods in the field's development, it is important to defend what can be learned from it and define what are the case's boundaries and scope. We have argued that the case is representative of many forms of tech-mediated communities– i.e., a “modal” case (DeSantola & Gulati, 2017; Hsieh & Vergne, 2023; Mollick, 2014; O'Mahony & Bechky, 2008). In the case of platforms, they too have dedicated technology communities of producers and consumers, a mediated relationship among them via the platform, and innovation coming in from projects and experiments. Hence, our modal case, in many ways, extends to them.

However, tech-mediated communities that just depend on a technology product, like a ham radio (Croidieu & Kim, 2018), or digital services (Santos & Eisenhardt, 2009), depend less on the platform and its core community than on start-up organizations and their founders. As a result, there is more early competition for a key “prototype” or “design” that enables the mechanism of becoming a “cognitive referent” (Santos & Eisenhardt, 2009) in addition to establishing boundaries. Phase 1 in our model would need modification to account for these elements. Phase 2 is closer to extant start-up resourcing and hence more applicable to those cases and adds to our knowledge of them; but Phase 3 on the reframing and language/philosophy change, then division of labor elaboration for a logic shift is more similar to work on scaling up communities (Howard-Grenville et al., 2017) than technology firms. That shift in potential representativeness is interesting to note. In future research using multiple technology startup or platform cases to examine Fig. 3's process model elements would be valuable.

Another important limitation is around our ability to capture co-constitution fully. Co-constitution implies the back-and-forth of macro-and-micro, especially of logics & ideologies in the local field with technology affordances, during scaling up (Faik et al., 2020; Lounsbury, Steele, Wang, & Toubiana, 2021; Schildt, 2020). In this study, our historical analysis of archival material in the empirical setting section of the methods detailed the two local field logics that were being enacted to varying degrees during Phase 1 and Phase 2 of Ethereum's rise. We also noted topics, such as topic 29- that entailed the technology community democratic logic- and its rise and fall, over time in the field in Fig. 2. But it was difficult to track the community and market logics throughout the seven-year history in the detail that is needed to provide a fully convincing co-constitutive analysis. Furthermore, the commentary from technology community and field members about sustainability and financial due diligence in Phase 3, while suggestive of a new, blended logic that might be emerging locally, was not something we were able to tie carefully to the topic modeling and content data. We do believe that such a logic was rising in the era (also see Yan et al., 2021) and being developed in regulatory contests around the FTX crypto scandal affecting the field. If that is so, pushing our current study forward with Ethereum and other blockchain platforms would be a valuable endeavor.

Credit authorship contribution statement

Yunjung Pak: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Timothy R. Hannigan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **P. Devereaux Jennings:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Michael Lounsbury:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Conceptualization.

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Appendix A. Supplementary data

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Data availability

Data will be made available on request.

References

- Almandoz, J. (2017). Drivers of community strength: an institutional logics perspective on geographical and affiliation-based communities. In *The SAGE handbook of organizational institutionalism* (p. 190).
- Anteby, M. (2010). Markets, morals, and practices of trade: Jurisdictional disputes in the US commerce in cadavers. *Administrative Science Quarterly*, 55(4), 606–638.
- Audia, P. G., Freeman, J. H., & Reynolds, P. D. (2006). Organizational foundations in community context: Instruments manufacturers and their interrelationship with other organizations. *Administrative Science Quarterly*, 51(3), 381–419.
- Augustine, G., Soderstrom, S., Milner, D., & Weber, K. (2019). Constructing a distant future: Imaginaries in geoeengineering. *Academy of Management Journal*, 62(6), 1930–1960.
- Barley, S. R. (1986). Technology as an occasion for structuring: Evidence from observations of CT scanners and the social order of radiology departments. *Administrative Science Quarterly*, 31(1), 78–108.
- Barnett, W. P. (1990). The organizational ecology of a technological system. *Administrative Science Quarterly*, 31–60.
- Bechky, B. A. (2003). Sharing meaning across occupational communities: The transformation of understanding on a production floor. *Organization Science*, 14(3), 312–330.
- Benford, R. D., & Snow, D. A. (2000). Framing processes and social movements: An overview and assessment. *Annual Review of Sociology*, 26(1), 611–639.
- Bohn, S., & Rogge, J. C. (2022). The framing of green innovations—A comparative topic modeling study on the public frames of the electric vehicle in Germany and UK. *Journal of Cleaner Production*, 364, Article 132499.
- Brody, A., & Couture, S. (2021). Ideologies and imaginaries in blockchain communities: The case of Ethereum. *Canadian Journal of Communication*, 46(3), 543–561.
- Buterin, V. (2014). *A next-generation smart contract and decentralized application platform*. Ethereum Foundation.
- Buterin, V. (2017). The meaning of decentralization. Medium. February 6 <https://medium.com/@VitalikButerin/the-meaning-of-decentralization-a0c92b76a274>.
- Buterin, V. (2023, December 28). “Make Ethereum Cypherpunk Again” Retrieved from <https://vitalik.eth.limo/general/2023/12/28/cypherpunk.html>.
- Caliskan, K. (2020). Data money: The socio-technical infrastructure of cryptocurrency blockchains. *Economy and Society*, 49(4), 540–561.
- CoinDesk. (2021). Valid points: More Ethereum upgrades to come after proof of stake, Buterin says. *CoinDesk*. April 28. Retrieved from <https://www.coindesk.com/tech/2021/04/28/valid-points-more-ethereum-upgrades-to-come-after-proof-of-stake-buterin-says/>.
- Croidieu, G., & Kim, P. H. (2018). Labor of love: Amateurs and lay-expertise legitimation in the early US radio field. *Administrative Science Quarterly*, 63(1), 1–42.
- Dahlander, L., & Frederiksen, L. (2012). The core and cosmopolitans: A relational view of innovation in user communities. *Organization Science*, 23(4), 988–1007.
- Davis, G. F., & Kim, S. (2015). Financialization of the economy. *Annual Review of Sociology*, 41(1), 203–221.
- Davis, G. F., & McAdam, D. (2000). Corporations, classes, and social movements after managerialism. *Research in Organizational Behavior*, 22, 193–236.
- Davis, J. L., & Chouinard, J. B. (2016). Theorizing affordances: From request to refuse. *Bulletin of Science, Technology & Society*, 36(4), 241–248.
- DeSantola, A., & Gulati, R. (2017). Scaling: Organizing and growth in entrepreneurial ventures. *Academy of Management Annals*, 11(2), 640–668.
- DiMaggio, P., Nag, M., & Blei, D. (2013). Exploiting affinities between topic modeling and the sociological perspective on culture: Application to newspaper coverage of US government arts funding. *Poetics*, 41(6), 570–606.
- Dodd, N. (2018). The social life of bitcoin. *Theory, Culture and Society*, 35(3), 35–56. <https://doi.org/10.1177/0263276417746464>
- DuPont, Q. (2017). Experiments in algorithmic governance: A history and ethnography of “The DAO,” a failed decentralized autonomous organization. In *Bitcoin and beyond* (pp. 157–177). Routledge.
- DuPont, Q. (2019). *Cryptocurrencies and blockchains*. John Wiley & Sons.
- Durkheim, E. (1893/2014). *The division of labor in society*. Simon and Schuster.
- Dylan-Ennis, P. (2024). *Absolute essentials of Ethereum (1st edition)*. Routledge.
- Dylan-Ennis, P., Kavanagh, D., & Araujo, L. (2023). The dynamic imaginaries of the Ethereum project. *Economy and Society*, 52(1), 87–109.
- Ethereum Foundation. (2024a). *The merge*. Ethereum Foundation. Retrieved from <https://ethereum.org/en/roadmap/merge/>.
- Ethereum Foundation. (2024b). *The Beacon*. Ethereum Foundation. Retrieved from <https://ethereum.org/en/roadmap/beacon-chain/>.
- Ethereum Foundation. (2025). ERC-721 Non-Fungible Token Standard. Retrieved from <https://ethereum.org/developers/docs/standards/tokens/erc-721/>.
- Faik, I., Barrett, M., & Oborn, E. (2020). How information technology matters in societal change: An affordance-based institutional logics perspective. *MIS Quarterly*, 44(3).
- Faraj, S., Jarvenpaa, S. L., & Majchrzak, A. (2011). Knowledge collaboration in online communities. *Organization Science*, 22(5), 1224–1239. <https://doi.org/10.1287/orsc.1100.0614>
- Feller, J., Finnegan, P., Fitzgerald, B., & Hayes, J. (2008). From peer production to productization: A study of socially enabled business exchanges in open source service networks. *Information Systems Research*, 19(4), 475–493.
- Furnari, S. (2016). Institutional fields as linked arenas: Inter-field resource dependence, institutional work and institutional change. *Human Relations*, 69(3), 551–580.
- Garud, R., & Karnøe, P. (2003). Bricolage versus breakthrough: Distributed and embedded agency in technology entrepreneurship. *Research Policy*, 32(2), 277–300.
- Garud, R., & Rappa, M. A. (1994). A socio-cognitive model of technology evolution: The case of cochlear implants. *Organization Science*, 5(3), 344–362.
- Gawer, A., & Phillips, N. (2013). Institutional work as logics shift: The case of Intel’s transformation to platform leader. *Organization Studies*, 34(8), 1035–1071.
- Giddens, A. (1991). *Modernity and self-identity*. Cambridge.
- Graeber, D. (2002). The new anarchists. *New Left Review*, 13, 61–73.
- Greenwood, R., Raynard, M., Kodeih, F., Micelotta, E. R., & Lounsbury, M. (2011). Institutional complexity and organizational responses. *Academy of Management Annals*, 5(1), 317–371.
- Greenwood, R., Suddaby, R., & Hinings, C. R. (2002). Theorizing change: The role of professional associations in the transformation of institutionalized fields. *Academy of Management Journal*, 45(1), 58–80.
- Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). *Text as data: A new framework for machine learning and the social sciences*. Princeton University Press.
- Grodal, S. (2018). Field expansion and contraction: How communities shape social and symbolic boundaries. *Administrative Science Quarterly*, 63(4), 783–818.
- Grodal, S., & O’Mahony, S. (2017). How does a grand challenge become displaced? Explaining the duality of field mobilization. *Academy of Management Journal*, 60(5), 1801–1827.
- Gupta, V. (2015). *The Ethereum launch process*. Ethereum Foundation. Retrieved from <https://blog.ethereum.org/2015/03/03/ethereum-launch-process>.
- Gurses, K., & Ozcan, P. (2015). Entrepreneurship in regulated markets: Framing contests and collective action to introduce pay TV in the US. *Academy of Management Journal*, 58(6), 1709–1739.

- Hannigan, T. R., Briggs, A. R., Valadao, R., Seidel, M.-D. L., & Jennings, P. D. (2021). A new tool for policymakers: Mapping cultural possibilities in an emerging AI entrepreneurial ecosystem. *Research Policy*, 104315.
- Hannigan, T. R., & Casasnovas, G. (2020). New structuralism and field emergence: The co-constitution of meanings and actors in the early moments of social impact investing. In *Macrofoundations: Exploring the institutionally situated nature of activity* (pp. 147–183). Emerald Publishing Limited.
- Hannigan, T. R., Haans, R. F., Vakili, K., Tchaljian, H., Glaser, V. L., Wang, M., I. ... Jennings, P. D. (2019). Topic modeling in management research: Rendering new theory from textual data. *Academy of Management Annals*, 13(2), 586–632.
- Hannigan, T. R., Pak, Y., & Jennings, P. D. (2022). Mapping the multiverse: A cultural cartographic approach to realizing entrepreneurial possibilities. *Research in the Sociology of Organizations*, 80, 239–259.
- Hargadon, A. B., & Bechky, B. A. (2006). When collections of creatives become creative collectives: A field study of problem solving at work. *Organization Science*, 17(4), 484–500.
- Hargadon, A. B., & Douglas, Y. (2001). When innovations meet institutions: Edison and the Design of the Electric Light. *Administrative Science Quarterly*, 46(3), 476–501.
- Hayes, A. (2019). The socio-technological lives of bitcoin. *Theory, Culture and Society*, 36(4), 49–72.
- Hedberg, L., & Lounsbury, M. (2020). Not just small potatoes: Cultural entrepreneurship in the moralizing of markets. *Organization Science*, 32(2), 433–454.
- Hinings, B., Logue, D., & Zietsma, C. (2017). Fields, institutional infrastructure and governance. In R. Greenwood, C. Oliver, T. B. Lawrence, & R. E. Meyer (Eds.), *The SAGE handbook of organizational institutionalism* (pp. 170–197). London: SAGE Publications Ltd.
- Howard-Grenville, J., Nelson, A. J., Earle, A. G., Haack, J. A., & Young, D. M. (2017). “If chemists don’t do it, who is going to?” peer-driven occupational change and the emergence of green chemistry. *Administrative Science Quarterly*, 62(3), 524–560.
- Hsieh, Y. Y., & Vergne, J. P. (2023). The future of the web? The coordination and early-stage growth of decentralized platforms. *Strategic Management Journal*, 44(3), 829–857.
- Hsieh, Y.-Y., Vergne, J.-P., Anderson, P., Lakhani, K., & Reitzig, M. (2018). Bitcoin and the rise of decentralized autonomous organizations. *Journal of Organization Design*, 7(1), 14.
- Jasanoff, S. (2015). Future imperfect: Science, technology, and the imaginations of modernity. In S. Jasanoff, & S.-H. Kim (Eds.), *Dreamscapes of modernity: Sociotechnical imaginaries and the fabrication of power* (pp. 1–33). University of Chicago Press.
- Jasanoff, S., & Kim, S.-H. (2015). Social movements and contested sociotechnical imaginaries in South Korea. In S. Jasanoff, & S.-H. Kim (Eds.), *Dreamscapes of modernity: Sociotechnical imaginaries and the fabrication of power* (pp. 152–33). University of Chicago Press.
- Jones, C., Boxenbaum, E., & Anthony, C. (2013). The immateriality of material practices in institutional logics. In *Institutional logics in action, part a* (pp. 51–75). Emerald Group Publishing Limited.
- Kellogg, K. C. (2009). Operating room: Relational spaces and microinstitutional change in surgery. *American Journal of Sociology*, 115(3), 657–711.
- Kennedy, M. T. (2008). Getting counted: Markets, media, and reality. *American Sociological Review*, 73(2), 270–295.
- Kim, S., & Kim, A. (2022). Going viral or growing like an oak tree? Towards sustainable local development through entrepreneurship. *Academy of Management Journal*, 65(5), 1709–1746.
- Kinney, A. B. (2020). The extra-institutional logics of an emerging market field: Values and practice in the bitcoin experience. *Social Currents*, 7(2), 109–130.
- Klein, J., & Amis, J. M. (2021). The dynamics of framing: Image, emotion, and the European migration crisis. *Academy of Management Journal*, 64(5), 1324–1354.
- Kraatz, M. S., & Block, E. S. (2017). Institutional pluralism revisited. In R. Greenwood, C. Oliver, T. B. Lawrence, & R. E. Meyer (Eds.), *The SAGE handbook of organizational institutionalism* (pp. 532–557). London: SAGE Publications Ltd.
- Lee, M. D. P., & Lounsbury, M. (2015). Filtering institutional logics: Community logic variation and differential responses to the institutional complexity of toxic waste. *Organization Science*, 26(3), 847–866.
- Lehdonvirta, V., & Robleh, A. (2016). *Governance and regulation. Distributed ledger technology: Beyond Blockchain*. London: UK Government Office for Science.
- Leonardi, P. M. (2011). When flexible routines meet flexible technologies: Affordance, constraint, and the imbrication of human and material agencies. *MIS Quarterly*, 147–167.
- Leonardi, P. M., & Barley, S. R. (2010). What’s under construction here? Social action, materiality, and power in constructivist studies of technology and organizing. *Academy of Management Annals*, 4(1), 1–51.
- Leonardi, P. M., & Vaast, E. (2017). Social media and their affordances for organizing: A review and agenda for research. *Academy of Management Annals*, 11(1), 150–188.
- Litrico, J. B., & David, R. J. (2017). The evolution of issue interpretation within organizational fields: Actor positions, framing trajectories, and field settlement. *Academy of Management Journal*, 60(3), 986–1015.
- Lounsbury, M. (2007). A tale of two cities: Competing logics and practice variation in the professionalizing of mutual funds. *The Academy of Management Journal*, 50(2), 289–307. <https://doi.org/10.2307/20159855>
- Lounsbury, M., & Glynn, M. A. (2019). *Cultural entrepreneurship: A new agenda for the study of entrepreneurial processes and possibilities*. Cambridge University Press.
- Lounsbury, M., Steele, C. W. J., Wang, M. S., & Toubiana, M. (2021). New directions in the study of institutional logics: From tools to phenomena. *Annual Review of Sociology*, 47(1), 261–280.
- Lounsbury, M., & Wang, M. S. (2020). Into the clearing: Back to the future of constitutive institutional analysis. *Organization Theory*, 1(1) (2631787719891173).
- Marquis, C., & Lounsbury, M. (2007). Vive La résistance: Competing logics and the consolidation of U.S. community banking. *Academy of Management Journal*, 50(4), 799–820.
- Marquis, C., Lounsbury, M., & Greenwood, R. (2011). Introduction: Community as an institutional order and a type of organizing. In , Vol. 33. *Communities and organizations*. Emerald Group Publishing Limited. pp. ix–xxvii.
- Massa, F. G. (2017). Guardians of the internet: Building and sustaining the anonymous online community. *Organization Studies*, 38(7), 959–988.
- Meyer, R. E., & Höllerer, M. A. (2010). Meaning structures in a contested issue field: A topographic map of shareholder value in Austria. *Academy of Management Journal*, 53(6), 1241–1262.
- Mohr, J. W., Bail, C. A., Frye, M., Lena, J. C., Lizardo, O., McDonnell, T. E., ... Wherry, F. F. (2020). *Measuring culture*. Columbia University Press.
- Mollick, E. (2014). The dynamics of crowdfunding: An exploratory study. *Journal of Business Venturing*, 29(1), 1–16.
- Murray, A., Kuban, S., Josefy, M., & Anderson, J. (2021). Contracting in the smart era: The implications of blockchain and decentralized autonomous organizations for contracting and corporate governance. *Academy of Management Perspectives*, 35(4), 622–641.
- News, F. (2020). *How Ethereum classic’s 51% attacks reveal risks to bitcoin and Ethereum*. Forkast News. September 7. (Retrieved from <https://forkast.news/video-achusterereum-classic-repeat-hacks-etc-labs-ceo-terry-culver-ben-sauter/>).
- O’Mahony, S., & Bechky, B. A. (2008). Boundary organizations: Enabling collaboration among unexpected allies. *Administrative Science Quarterly*, 53(3), 422–459.
- O’Mahony, S., & Ferraro, F. (2007). The emergence of governance in an open source community. *Academy of Management Journal*, 50(5), 1079–1106.
- Orlikowski, W. J., & Barley, S. R. (2001). Technology and institutions: What can research on information technology and research on organizations learn from each other? *MIS Quarterly*, 145–165.
- Pache, A. C., & Santos, F. (2013). Inside the hybrid organization: Selective coupling as a response to competing institutional logics. *Academy of Management Journal*, 56(4), 972–1001.
- Pak, Y., Hannigan, T., & Jennings, P. D. (2023). Crisis and institutional voids: The unintended normative consequence of multiverse moves in response to the ICO blockchain field hack. In *Proceedings of the European group for organizational studies conference, Cagliari*.
- Peters, T. (1983). The mythology of innovation, or a skunkworks tale, part II. In M. L. Tushman, & W. L. Moore (Eds.), *Readings in the management of innovation* (Reprinted in, pp. 138–147). Cambridge, MA: Ballinger.
- Powell, W. W., White, D. R., Koput, K. W., & Owen-Smith, J. (2005). Network dynamics and field evolution: The growth of interorganizational collaboration in the life sciences. *American Journal of Sociology*, 110(4), 1132–1205.

- Rameder, H., di Angelo, M., & Salzer, G. (2022). Review of automated vulnerability analysis of smart contracts on Ethereum. *Frontiers in Blockchain*, 5. <https://doi.org/10.3389/fbloc.2022.814977>
- Reinecke, J. (2018). Social movements and prefigurative organizing: Confronting entrenched inequalities in occupy London. *Organization Studies*, 39(9), 1299–1321.
- Ries, E. (2011). *The lean startup: How today's entrepreneurs use continuous innovation to create radically successful businesses*. Crown Currency.
- Roberts, M. E., Stewart, B. M., & Tingley, D. (2015). Stm: An R package for structural topic models. *Journal of Statistical Software*, 91(1), 1–40.
- Rothschild-Whitt, J. (1979). The collectivist organization: An alternative to rational-bureaucratic models. *American Sociological Review*, 509–527.
- Santos, F. M., & Eisenhardt, K. M. (2009). Constructing markets and shaping boundaries: Entrepreneurial power in nascent fields. *Academy of Management Journal*, 52(4), 643–671.
- Saxenian, A. (1994). *Culture and competition in Silicon Valley and route 128*. Cambridge: Harvard University Press.
- Schildt, H. (2020). *The data imperative: How digitalization is reshaping management, organizing, and work*. USA: Oxford University Press.
- Schmiedel, T., Müller, O., & Vom Brocke, J. (2019). Topic modeling as a strategy of inquiry in organizational research: A tutorial with an application example on organizational culture. *Organizational Research Methods*, 22(4), 941–968.
- Scott, W. R. (2013). *Institutions and organizations: Ideas, interests, and identities*. Sage publications.
- Seidel, M.-D. L. (2018). Questioning centralized organizations in a time of distributed trust. *Journal of Management Inquiry*, 27(1), 40–44.
- Seidel, M. D. L., & Greve, H. R. (2017). Emergence: How novelty, growth, and formation shape organizations and their ecosystems. In *Emergence* (pp. 1–27). Emerald Publishing Limited.
- Serada, A., Sihvonen, T., & Harviainen, J. T. (2021). CryptoKitties and the new ludic economy: How blockchain introduces value, ownership, and scarcity in digital gaming. *Games and Culture*, 16(4), 457–480.
- Shane, S., & Venkataraman, S. (2001). Entrepreneurship as a field of research: A response to Zahra and Dess, Singh, and Erikson. *Academy of Management Review*, 26(1), 13–16.
- Simmel, G. (1898). The persistence of social groups. *American Journal of Sociology*, 3(5), 662–698.
- Srinivasan, B. (2022). *The network state: How to start a country*. <https://thenetworkstate.com/>.
- Swanson, E. B., & Ramiller, N. C. (1997). The organizing vision in information systems innovation. *Organization Science*, 8(5), 458–474.
- Swartz, L. (2018). What was bitcoin, what will it be? The techno-economic imaginaries of a new money technology. *Cultural Studies*, 32(4), 623–650.
- Thornton, P. H., & Ocasio, W. (1999). Institutional logics and the historical contingency of power in organizations: Executive succession in the higher education publishing industry, 1958–1990. *American Journal of Sociology*, 105(3), 801–843.
- Thornton, P. H., Ocasio, W., & Lounsbury, M. (2012). *The institutional logics perspective: A new approach to culture, structure and process*. OUP Oxford.
- Timmermans, S., & Tavory, I. (2012). Theory construction in qualitative research: From grounded theory to abductive analysis. *Sociological Theory*, 30(3), 167–186.
- Villamor Martin, M., Kirsch, D. A., & Prieto-Nañez, F. (2023). The promise of machine-learning-driven text analysis techniques for historical research: Topic modeling and word embedding. *Management & Organizational History*, 0(0), 1–16.
- Von Krogh, G., & Von Hippel, E. (2006). The promise of research on open source software. *Management Science*, 52(7), 975–983.
- Weick, K. E. (1976). Educational organizations as loosely coupled systems. *Administrative Science Quarterly*, 21(1), 1–19.
- Wellman, B. (1979). The community question: The intimate networks of east Yorkers. *American Journal of Sociology*, 84(5), 1201–1231.
- Westphal, J. D., Gulati, R., & Shortell, S. M. (1997). Customization or conformity? An institutional and network perspective on the content and consequences of TQM adoption. *Administrative Science Quarterly*, 366–394.
- Wry, T., Lounsbury, M., & Glynn, M. A. (2011). Legitimizing nascent collective identities: Coordinating cultural entrepreneurship. *Organization Science*, 22(2), 449–463.
- Wu, K., Ma, Y., Huang, G., & Liu, X. (2021). A first look at blockchain-based decentralized applications. *Software: Practice and Experience*, 51(10), 2033–2050.
- Yan, S., Almandoz, J., & Ferraro, F. (2021). The impact of logic (in) compatibility: Green investing, state policy, and corporate environmental performance. *Administrative Science Quarterly*, 66(4), 903–944.
- Yates, L. (2015). Rethinking prefiguration: Alternatives, micropolitics and goals in social movements. *Social Movement Studies*, 14, 1–21.
- Zachariadis, M., Hileman, G., & Scott, S. V. (2019). Governance and control in distributed ledgers: Understanding the challenges facing blockchain technology in financial services. *Information and Organization*, 29(2), 105–117. <https://doi.org/10.1016/j.infoandorg.2019.03.001>
- Zavolokina, L., Bauer-Hänsel, I., Hacker, J., & Schwabe, G. (2024). Organizing for value creation in blockchain information systems. *Information and Organization*, 34(3), Article 100522.
- Zietsma, C., Groenewegen, P., Logue, D. M., & Hinings, C. R. (2017). Field or fields? Building the scaffolding for cumulation of research on institutional fields. *Academy of Management Annals*, 11(1), 391–450.